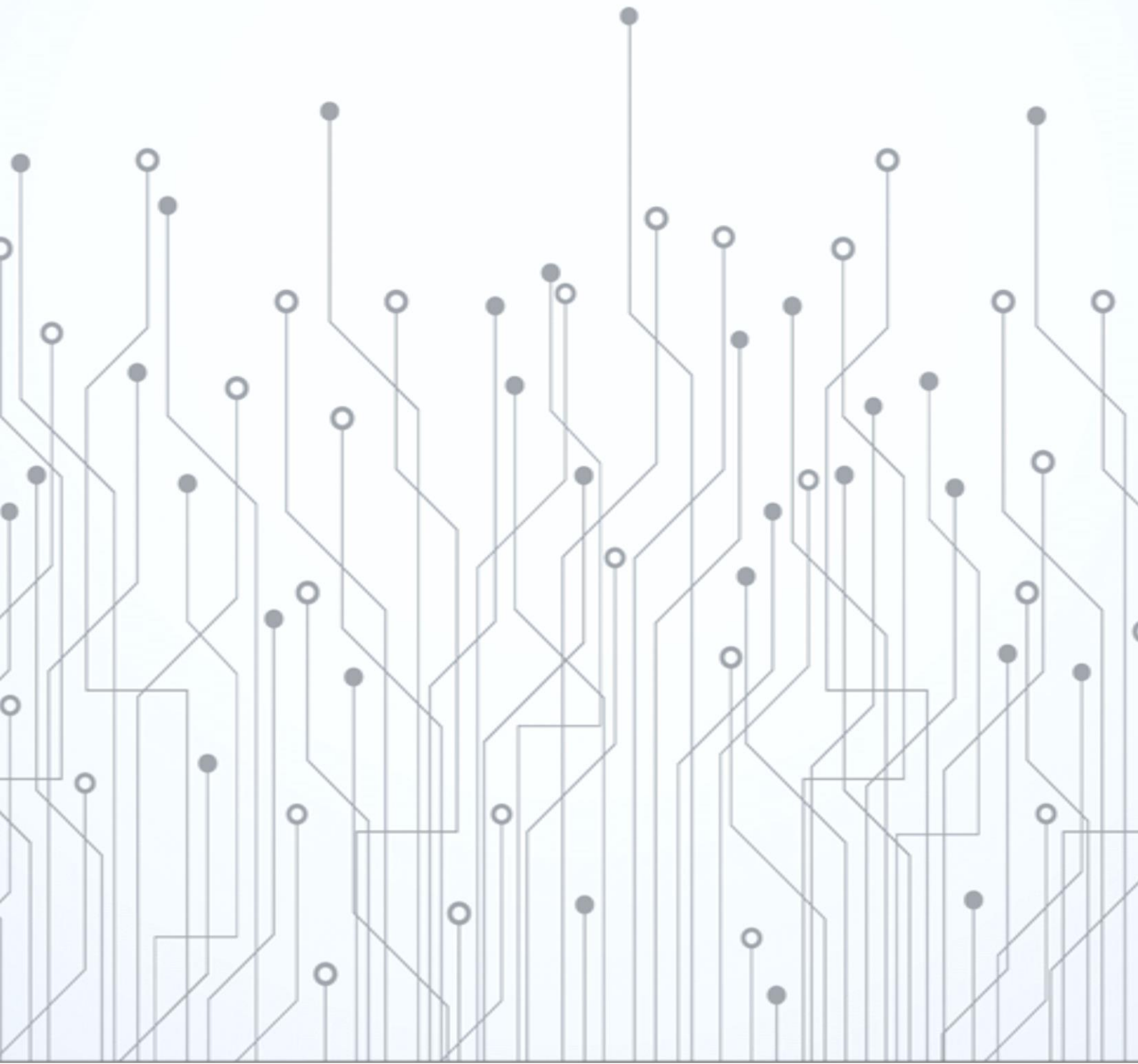




TECHNICAL GUIDELINES FOR CARTOGRAPHY

V1.0

General Authority for Survey and Geospatial Information



Disclaimer

This document has been developed by the General Authority for Survey and Geospatial Information (GEOSA). The guidelines have been drafted with reference to various standards and guidelines considered to be international best practices, particularly: the National Geospatial-Intelligence Agency (NGA) and the United States Geological Survey (USGS). Any omissions or errors may please be notified to GEOSA.

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Foreword

The General Authority for Survey and Geospatial Information (GEOSA) is the national mapping agency of the Kingdom of Saudi Arabia (KSA). These technical guidelines are a vital framework that align the Kingdom's geospatial data management practices with international and national standards. This will help industry, academia, and government bodies in KSA with survey and geospatial data collection and dissemination based on international best practice. The geospatial technical guidelines are derived from International and National Standards and will then be used to further develop guidelines and standard operating procedures (SOPs) for various geospatial and remote sensing data forms. This structured approach ensures that geospatial data collected and managed in KSA adhere to international and national standards, leading to data consistency, reliability, and compatibility.

GEOSA's mandate centers around the production and marketing of geospatial survey information and services to meet user and decision-maker requirements. Part of this mandate is to produce various map products, as an output of geodetic, topographic, or hydrographic surveys.

The guideline provides a framework for the creation of cartographic products in the form of digital map products comprising of map content, title, graticules, legend, and other text and symbols. As such, this guideline defines traditional map elements, spatial extents, spatial referencing systems, map sheet layout, cartographic processes, data quality, and data product delivery.

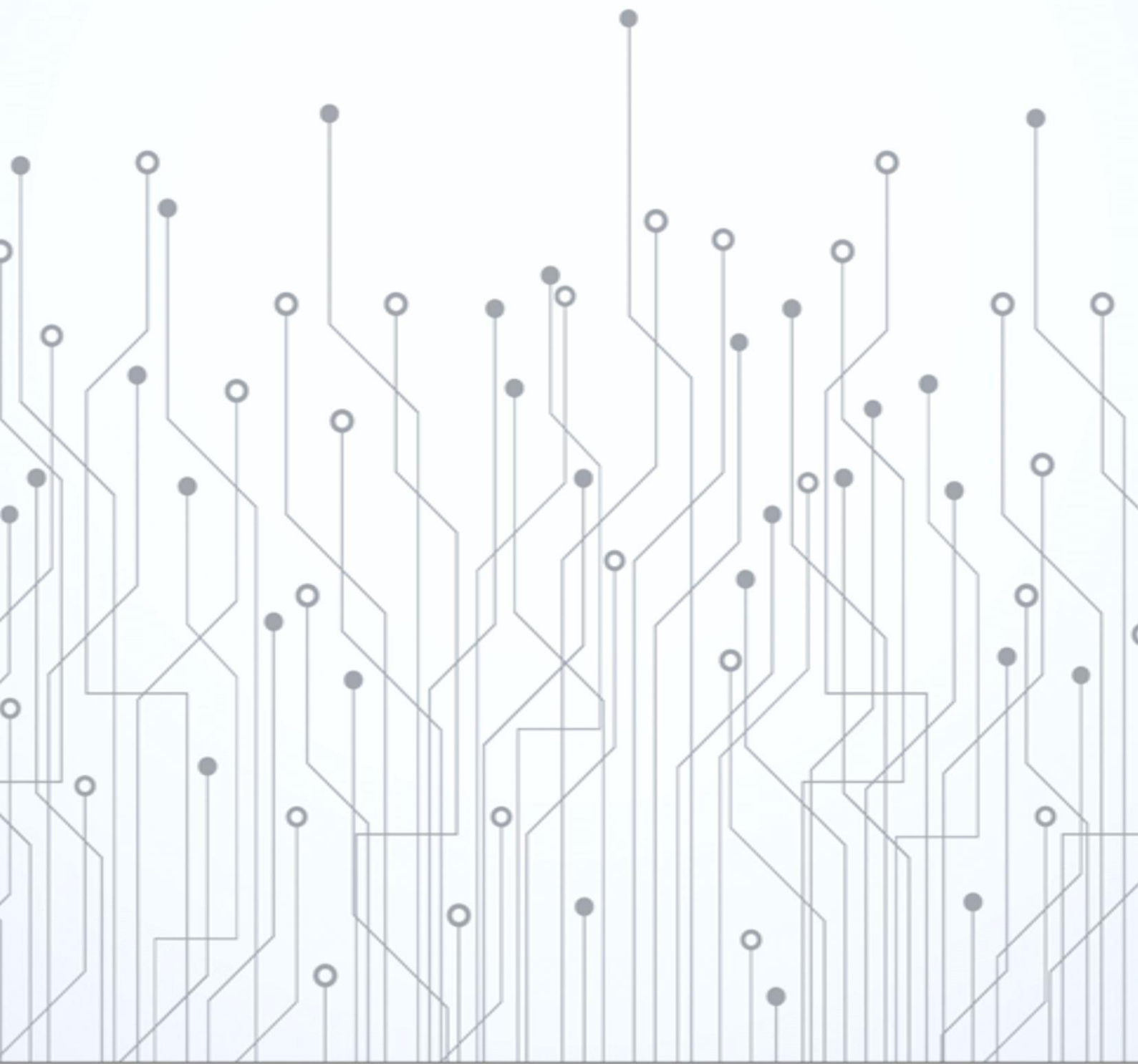
The guideline recognizes that, in addition to the traditional map series that exists in the Kingdom, map design and cartographic products are dynamic and created to suit the purpose of the map, sometimes for a specific use case. Detail, such as symbolization, is defined to suit the use case and intended scale or level of detail required. The guideline also recognizes the rapidly developing requirements to move from the traditional cartographic map sheet to the need for digital cartographic products, to be used for digital portrayal and which are not always bound by defined sheet lines.

The guideline provides general cartographic guidelines that are scale independent. Annexes are provided as containers intended for Standard Operating Procedures (SOP's) detailing the portrayal of the various map series products and other specialized cartographic products. The data portrayal discussed in this guideline is a framework to support web cartography.

The Guidelines for Cartography were developed in alignment with the National Geospatial Data Policies and Standards for the Kingdom of Saudi Arabia. Other technical guidelines in this series, including the Guidelines for Geodesy and Guidelines for Remote Sensing, have been referred to where found relevant in this document.

Changes to this Guidelines for Cartography will be instructed by GEOSA, as improvements or amendments are required.

1. Introduction



Scope and Objectives of the Guideline

The scope of this version of the KSA Cartography Guidelines is for the production of topographic maps in digital formats for the Kingdom of Saudi Arabia. The guidelines for thematic maps, such as land use maps, are not included.

Details relating to the creation & storage of topographic features, orthophoto production/sourcing, and color printing/plotting are excluded.

Terms and Definitions

Table 1: Terms and Definitions

Term	Definition
Data Product	A geospatial data product Guidelines is a precise technical description that characterizes a geospatial data product. It includes general information for data identification as well as information on data content and structure, reference systems, data quality aspects, data capture, maintenance, delivery, and metadata.
Dataset	Identifiable collection of data
Dataset Series	Collection of datasets sharing the same product Guidelines
Geospatial PDF	Portable Document Format (PDF) with a non-patented georeferencing extension
Metadata information	An information set about a map or other geospatial product that describes how the product was made and gives the sources of data and other relevant information

2. Normative Elements

Cartography Technical Guidelines



2.1 Background

As the leading national organization in topographic survey, mapping, charting, geographical information and hydrographic survey in the Kingdom of Saudi Arabia, the General Authority for Survey and Geospatial Information (GEOSA) mandate centers around the production and marketing of geospatial survey information and services, improving relevant technologies, high-quality training and development of manpower, and cooperation with GEOSA partners to meet user and decision-maker requirements.

The GEOSA mandate is to carry out geodetic, topographic, and hydrographic surveys, as well as map production, followed by the development of Geoinformation systems that are relevant to GEOSA activities.

This guideline is for defining KSA topographic map products at various scales. GEOSA has in the past produced map series products at defined scales. This guideline does not limit cartographic products to any particular map series but provides a framework that is scale independent. Annexures are provided as containers for Standard Operating Procedures (SOP's) for feature catalogs, portal catalogs, annotation catalogs and style sheets to suit particular use cases or standardized map series products.

The guideline provides a framework for the creation of a cartographic product in the form of a traditional map sheet containing map content, title, graticules, legend, and other text and symbols.

The Guideline is based on the frameworks proposed by relevant ISO Standards, namely:

- ISO 19131:2022 – Geographic information Data product guideline (DPS).
- ISO 19117:2012– Geographic information Portrayal.

The ISO 19131, Geographic information - Data Product Guideline defines a standard format and structure for guidelines for geospatial data products. The standard also considers a hardcopy map or chart to be a data product.

Content and application of the ISO frameworks have been adopted from the Cartography Guidelines of the United States Geological Survey (USGS) and US National Geospatial-Intelligence Agency (NGA) and adapted to suit the requirements of the Kingdom of Saudi Arabia, for example by accommodating Arabic text and sheet naming requirements.

Annotation, Portrayal and Style Sheet guidelines are NGA compliant examples to serve as a basis for further GEOSA enhancement of symbolization, annotation and text.

2.1.1 Applicability of the Guideline

The KSA Cartography Guidelines are intended to serve conventional map users by providing geographic information system (GIS) information in a symbolized form, in the customary topographic map layout. The maps are not intended for advanced GIS analysis applications. These products are built on standard coordinate systems and used by traditional topographic map users, such as resource managers, planners, emergency first responders, and recreational users who continue to need the symbolized feature data.

Full-size Style Sheet example templates in Portable Document Format (PDF) defining the placement of map elements, marginalia, font sizes, and styles also accompany this guideline. KSA topographic maps published in geospatial PDF and other formats are fashioned to conform to these style sheets so that a user can print out a map. For example, at the 1:25,000 scale, using the dimensions of the traditional standard 7.5-minute quadrangle.

Allowance is made for variations to a standard sheet containing only a cartographic representation of topographical features. This includes map variations containing orthophoto imagery and/or sun-shaded relief detail.

The creation of ortho imagery, whether from satellite or aerial sources, is not included in this guideline. Similarly, the methods of printing a traditional hardcopy paper map sheet, whether by inkjet plotting or four colors - cyan, magenta, yellow, and black (CMYK) lithographic printing process are not covered by this guideline as these processes fall under Standard Operation Procedures (SOP's) to be documented elsewhere.

This Guideline does not cover the creation of the topographic features required for the map content as these data are available in the GEOSA National Topographic Database as defined by the KSA National Feature Concept Dictionary (KSA-NFCD)() and the Topographic Data Store (TDS). There are existing procedures and guidelines in GEOSA for the creation and updating of topographic map data using the Topographic Data Store Extraction Guide (TDS EG) and Topographic Data Store Entity Catalog (TDS EC).

The KSA topographic map is a large-scale map characterized by a high level of planimetric detail and quantitative representation of relief using elevation contour lines. The various features shown on the map are represented by standard symbols. These symbols are explained in the margin of the map along with other information about the map and its content.

Topographic map content includes symbolization of transportation and cultural features, vertical obstructions, hydrography, hypsography, vegetation, boundaries, and geographic- placenames. Some instances of map use may include a Military Grid Reference System (MGRS) grid.

Examples of marginalia annotation include a map legend to illustrate and define symbols used on the map, a glossary of geographic terms, adjoining sheets diagram, boundaries diagram, bar scales, notes on a geographic reference system, datum, elevation contour interval, and sheet identification.

All units of measurement (elevations, heights, lengths, widths) are in meters unless otherwise noted.

Although primarily a “paper” product, a Topographic Map may also be converted to digital “copies” of paper maps, such as Portable Document Format (PDF).

The Guideline provides for a geospatial extension, in an International Organization for Standardization (ISO) 32000-compliant Adobe® PDF that is called a “Geospatial PDF.”

The traditional KSA Topographic Map Series are updated on a three-year revision cycle for urban areas and a five-year revision cycle for desert areas. Newly published maps as well as previously published, non-current, KSA topographic maps are available from info@geosa.gov.sa

This Guideline provides Annexures that represent containers for content required for Feature Catalog (FC), Annotation Catalog (AC), Portrayal Catalog (PC) and Style Sheets. Examples following the NGA standard are provided.

2.1.2 Guiding Principles

The Guiding Principles defining KSA Technical Guidelines require them to be Discoverable, Accessible, Interoperable, Repeatable, Authoritative, User-centric, and Extensible. These guiding principles are central to the compilation of this guideline document and have been used throughout the document.

The User-Centric requirement is paramount to ensuring the adoption and use of technical guidelines that are fit-for-purpose and meet user needs and use cases that exist within GEOSA and in the wider KSA Geospatial Community.

The Technical Guidelines are authoritative and extensible through means of publishing and ongoing maintenance and extensions by the GEOSA, an authoritative national mapping agency for the Kingdom of Saudi Arabia. This shall support the guiding principle of being extensible.

2.1.3 Need for KSA Topographic Maps

In the geospatial community, there is a need for general-purpose maps that can be easily printed at standard map scales. Similarly, there is a need for a national map series that can be used without specialized GIS software and expertise.

A topographic line map, with the optional addition of an ortho-image and shaded relief, is a valuable tool, particularly when provided in a digital format that allows the user to turn features and layers on and off, zoom in and out, and print the entire map or portions of the map at a standard map scale.

2.1.4 Additional Requirements

In addition to the need for usable cartographic data products, further needs, requirements, and objectives that must be met as part of this guideline are:

- Alignment with the National Geospatial Data Policies and Standards for the Kingdom of Saudi Arabia ([General Authority for Survey and Geospatial Information, GEOSA, 2023](#)).
- Align with the KSA – NFCD (National Feature Concept Dictionary) ([General Authority for Survey and Geospatial Information, GEOSA, 2022](#)) and Foundation Themes.
- Compatibility with existing cartographic workflows and software (e.g., ESRI Cartographic software workflows).
- Align with Multinational Geospatial Co-production Program (MGCP) requirements – if applicable or should this be a future requirement.
- Inclusion of Contours & Hill Shade (Shaded Relief) to provide visually rich cartographic products.
- Align with and source data from GEOSA National Geospatial Platform, National Topographic Database & Topographic Data Store (TDS).
- Need for a future-oriented view of cartographic data products that are created online and on demand.
- Need for ease of use and accessibility.
- The Map Sheet deliverables must be compiled to follow the examples in the following technical guideline Annexes:
 - As an example, the KSA Topographic Map Sheets (1:25,000 Scale) are to use a 7 minute, 30 seconds by 7 minute, 30 second, Geographic (Non-Projected) format extent, with a labeled 2000 meter spacing UTM projection grid overlay. See examples in : Topographic Map Guidelines Annotation Catalog (TMS AC).
 - Published KSA Topographic / Orthophoto Map Feature Symbolology will conform to the dimension, geometry, and color guidelines contained in : Topographic Map Guidelines Portrayal Catalog (TMS PC).
 - All Arabic text script diacritics and all transliterated Latin text on the published KSA topographic / Orthophoto Map deliverables, are to be reproduced as per the **Geographic Names Database** of GEOSA.
 - All map typefaces on the published KSA topographic / Orthophoto Map deliverables, will conform to the dimensions, geometry, and colors contained in : Topographic Map Guidelines Annotation Catalog (TMS AC).
 - Map sheet layouts on the published KSA topographic / Orthophoto Map deliverables, will conform to the layout details contained in : Topographic Map Guidelines Style Sheets (TMS SS).
 - Font encoding is to conform to the Unicode Register standard.
 - The map sheet Legend and Surround will conform to GEOSA supplied KSA topographic / Orthophoto Map Legend and Surround ESRI MXD (map document files) format legend templates.

2.2 Identification

This section provides an overview identification of the KSA topographic map as further specified in the remainder of this guideline document.

2.2.1 Abstract

Cartographic representation of digital map data is a requirement to meet many users' needs in the wider geospatial community in KSA. These can be cartographic products for the KSA national map series or other purpose designed cartography with scale and portrayal to suit a particular use case.

The 1:25,000 topographic maps are examples of digital and hardcopy maps used for civilian and military purposes in the Kingdom of Saudi Arabia. They portray transportation and cultural features, vertical obstructions, hydrography, hypsography, vegetation, boundaries, and geographic placenames, along with an (optional) Military Grid Reference System (MGRS) grid. Annotation in the margin of the map provides basic metadata about the information shown on the map, as well as additional non-feature related information, such as an Adjoining Sheets diagram, map legend, glossary, etc.

2.2.2 Title

Each KSA topographic map shall be identified by a sheet name or sheet number if forming part of a national map series. Maps covering vast desert areas are difficult to name due to the lack of prominent features or place names. In these cases, the map sheets are numbered. Each topographic map shall also be identified by a Reference Number (RN).

Sheet names and Reference Numbering shall be constructed according to the definitions and principles in [2.8 Data Product Identification](#)

2.2.3 Spatial Resolution

The spatial resolution for KSA topographic maps shall be at various scales to suit the use case.

2.2.4 Geographic Extent

In the case of a national map series, the geographic extent of the KSA topographic map sheet shall correspond to a guideline of sheet lines. This Guideline prescribes no positional limitations for a TM map sheet in terms of geographic location (valid throughout the Kingdom of Saudi Arabia).

For example, sheet lines for 1:25,000 scale topographic maps shall follow an established format as described in [Section 2.62.6 Map sheets](#).

2.2.5 Language

Two languages shall be employed in the creation of the KSA topographic map product. These shall be in English and Arabic. The selection of only English or only Arabic or the inclusion of another language may be considered for special projects and shall be specified in supplemental instructions for the project.

2.3 Spatial Extents

The guideline provides a final map product in the form of map sheets in print-ready PDF and other formats. For digital consumption or for plotting to a hardcopy map sheet.

Special cases may require an adaptation of the sheet extents.

See Annex E - Example 1:25 000 Scale Topographic Map

2.4 Spatial Reference System

2.4.1 KSA Geodetic Reference System

The **Saudi Arabia National Spatial Reference System (SANSRS)** is comprised of several geodetic components, which have been formally defined as a GML Spatial Reference System (SRS) Dictionary in the SANSRS Foundation Theme. The geodetic components consist of datums, coordinate systems (axes and units), and coordinate reference systems, which relate each coordinate system to a datum. These geodetic components are described in further detail in the sections that follow.

Any Dataset that is created or published according to this Data Guideline shall reference Universal Transverse Mercator (UTM) coordinate reference systems as defined in the Saudi Arabian National Spatial Reference System (SANSRS) Foundation Theme Data Guideline.

All maps are georeferenced to the KSA National Spatial Reference System, World Geodetic System of 1984 (WGS 84). The ellipsoid for KSA topographic maps shall be World Geodetic System 1984 (WGS 84). The horizontal datum for KSA topographic maps shall be World Geodetic System 1984 (WGS 84). The vertical datum for KSA topographic maps shall be Mean Sea Level (MSL) based on the Kingdom of Saudi Arabia Geoid (KSA-GEOID2017). Jeddah 1970 Height System - All heights should be expressed with reference to MSL Jeddah Tide Gauge, 1970. KSA Geoid (currently Saudi Arabia Geoid (KSA-GEOID2017) shall be used for transformations from ellipsoidal to orthometric heights.

2.4.2 Projection

Saudi Arabia spans multiple UTM zones due to its large geographical extent. The UTM system divides the world into 6-degree longitudinal zones, and each zone has a unique identification number. The UTM zones used in Saudi Arabia typically fall within Zones 37 to 40 as described below:

- **UTM Zone 37:** This zone extends from 36°E to 42°E and covers the northwestern part of Saudi Arabia, including cities like Tabuk and the border region with Jordan.
- **UTM Zone 38:** This zone spans from 42°E to 48°E and covers the central part of Saudi Arabia, including the capital city, Riyadh.
- **UTM Zone 39:** This zone stretches from 48°E to 54°E and covers the eastern and northeastern parts of Saudi Arabia, including cities like Dammam and Al Khobar.
- **UTM Zone 40:** This zone ranges from 54°E to 60°E and covers the southeastern part of Saudi Arabia, including the Asir region and the border with Yemen.

These UTM zones facilitate accurate map projections and coordinate systems for cartographic and geospatial purposes. When working with geographical data in Saudi Arabia, it's essential to determine the appropriate UTM zone for any specific location to ensure accurate and consistent measurements and mapping.

In particular circumstances it may be more appropriate to use the Lambert's Conical Conform Projection with Two Standards Parallels.

2.4.3 Temporal Reference Systems

The default temporal reference systems for KSA are the Gregorian and Islamic (Hijri) calendars, with dates expressed in accordance with ISO 8601 Data Elements and Interchange Formats – Information Interchange – Representation ([International Standards Organization, ISO, 2017](#)). ISO 8601 is an international standard covering the exchange of date and time-related data.

Any dataset that is created or published according to this data guideline shall express temporal references such as dates in accordance with ISO 8601 Data Elements and Interchange Formats – Information Interchange – Representation, of dates and times using both the Gregorian and Islamic (Hijri) calendars.

The purpose of this standard is to provide an unambiguous and well-defined method of representing dates and times, to avoid misinterpretation of numeric representations of dates and times, particularly when data is transferred between countries with different conventions for writing numeric dates and times. The standard organizes the data, so the largest temporal term (the year) appears first in the data string and progresses to the smallest term (the second). It also provides for a standardized method of communicating time-based information across time zones by attaching an offset to Coordinated Universal Time (UTC). EXAMPLE 1997 (the year 1997), 1997-07-16 (16th July 1997), 1997-07-16T19:20:30+01:00 (16th July 1997, 19h 20' 30", time zone: UTC+1).

2.5 Data Content and Structure

The KSA topographic maps are feature-based digital and hardcopy map products, based on the graphic symbolization of geospatial features contained within the map sheet limits.

In the case of the KSA topographic /Ortho Map, the Orthorectified Project Imagery Base will have the Project, Hydrographic, Transportation, Building, Contour, Hillshade, and Geographic Names Project Databases, cartographically overlaid.

2.5.1 Data Capture

Geospatial information portrayed on KSA topographic maps shall be taken from the KSA National Topographic Database Entity Catalog which includes boundaries, elevation, geographic names, topographic, maritime, and aeronautical detail.

2.5.2 Feature-based data

The KSA Topographic Map Guideline Feature Catalog (TMS FC) shall be a subset of the KSA National Topographic Datastore Entity Catalog consisting of those features relevant for portrayal on a topographic map. Features not listed in shall not be portrayed.

Features shall be portrayed based on the symbol criteria contained in the Topographic Map Guideline Portrayal Catalog (TMS PC). However, not every instance of a feature may be symbolized on a TM. Some feature geometries or combinations of feature attributes and enumerates are not to be portrayed.

See : Feature Catalog for the KSA topographic map Guideline Feature Catalog (TMS FC) for examples.

2.6 Map sheets

Sheet lines are how a geographic area is divided to establish the limits of individual sheets. Sheet lines shall generally be formed by parallels of latitude and meridians of longitude. The sheet lines of individual maps are also referred to as neatlines.

The use of geographic coordinates as a system of reference is based on the expression of position by latitude and longitude in degrees, minutes, and decimal minutes. The projection shall be represented on the map by limiting sheet lines (neatlines) and a series of evenly spaced projection intersections in the map interior.

For example, these intersections shall be at 2 minutes of arc intervals for the 1:25,000 scale and the sheet lines shall show the meridians (lines of longitude) as straight lines, and parallels (lines of latitude) which affect curvature through the connection of straight-line segments between successive intermediate projection intersections. The intersections of the parallels and meridians of the projection shall be plotted within 0.15 mm of the computed position.

The Universal Transverse Mercator (UTM) Grid shall be shown on the Topographic Maps.

The Military Grid Reference System (MGRS), if applicable, shall be used to express positions in terms of UTM grid coordinates.

The grid interval on topographic maps shall be determined according to the scale required in both northing and easting, for example 1000m in the case of 1:25,000 scale. The grid shall be constructed on a given sheet so that the distances between adjacent grid lines shall not vary more than 0.15 mm from the computed grid interval; and the overall distances between the first full grid lines, complementing those of adjoining sheets, shall not vary more than 0.15 mm from their computed measurements.

2.6.1 Off-Grid and Oversized Maps

In some cases, the extent of the KSA topographic maps does not precisely match the domain of a standard 7.5-minute cell (in the case of 1:25,000 scale). For example, for some maps, the projection line is extended to show all of the land mass or a water feature or to match a boundary. In extreme cases, the entire projection line is shifted, for example, to center the map over an island.

GEOSA cell-based digital products, such as digital elevation models (DEM), do not allow for such exceptions. These products are restricted to the geographic domain of an on-grid 7.5-minute cell.

A policy of not allowing oversize or off-grid maps has several implications:

- The development and maintenance of map-production software and procedures is simpler. This is the primary reason the current guideline dictates only on-grid, standard-size products. Building production systems to accommodate the special cases of off-grid products is difficult and the number of cells involved is small.
- Softcopy map users will probably not be significantly inconvenienced by this policy and may consider the higher level of regularity and standardization to be a benefit.
- The primary effect of this policy is on users who rely on paper maps. There will be some instances of maps that are covered almost completely by water and instances of small island chains broken up into several pieces. In these cases, users who print their maps will use more ink and paper to print very little additional information.

2.6.2 Establishment of standard sheet lines

The establishment of sheet lines is based on the following principles:

- Sheet lines shall be developed on a series or project basis.
- Sheet lines shall be designed to provide map coverage of an area with the minimum number of sheets without unduly impairing the continuity of adjoining sheets.
- Sheet lines shall be so positioned that they coincide with the grid, ellipsoid, and datum junctions wherever possible.
- Sheet lines may vary within a map series and between map scales.

Table 2 lists the standard 1:25,000 scale sheet line sizes and the latitudes applicable to KSA as examples:

Table 2: Example of Standard 1:25,000 Scale Sheet Line Sizes and Applicable Latitudes

Latitude	Sheet Sizes
	1:25,000
	N-S E-W
0° to 36°	7.5' x 7.5'

2.6.3 Departures from standard sheet lines

Certain departures from the standard sheet lines may be required to avoid unnecessary sheets, thereby reducing the number of map sheets in a project. However, these departures shall be kept at a minimum and shall be based on careful consideration of their impact on the overall requirement of continuity of standard sheet lines. Departures from standard sheet lines occur most often in coastal areas, long narrow islands, and large islands with varying widths.

Base considerations when addressing the introduction of departures from standard sheet lines should include the following principles and options:

- Adherence to normal preferred maximum work limits (see [Section 2.10.32.10.3 Work limits](#)).
- Extent of land topography and the need to show landmark coastal hydrographic features.
- The placement of margin data in open water areas.
- Existence of grid, ellipsoid, and datum junctions.
- Adjustment of sheet lines to avoid decimal parts of second-of-arc.

2.6.4 Examples of non-standard sheets

The following are examples of departures from standard sheet lines.

NOTE: the dashed lines in the figures represent where the standard sheet lines would exist if unadjusted.

2.6.4.1 Border break extensions

A border break extension permits an adjustment to a sheet neatline to accommodate islands of an adjoining area, small points of land, or offshore features (see Figure 1 and Figure 2

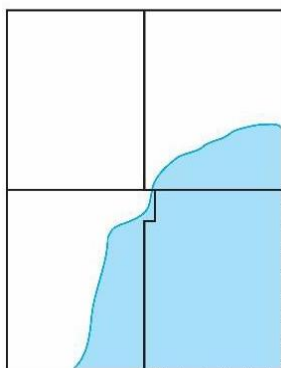


Figure 2Error! Reference source not found.). When there is a choice of sheets that may contain a border break extension, the sheet that requires the least re-arrangement of margin data shall be selected. The neatline is extended around the protruding land mass and incorporates all applicable features within.

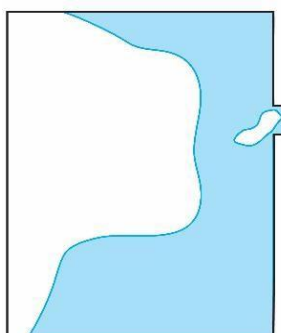


Figure 1: Border Break for Island

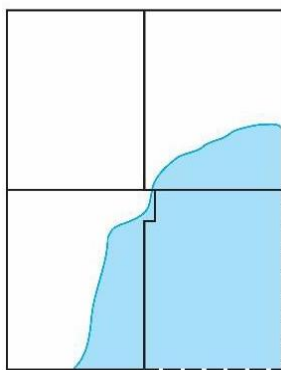


Figure 2: Border Break for Coastal Land

2.6.4.2 Extensions

An extension (Figure 3) is the enlargement of a sheet by moving one or more sheet lines to include adjoining land areas.

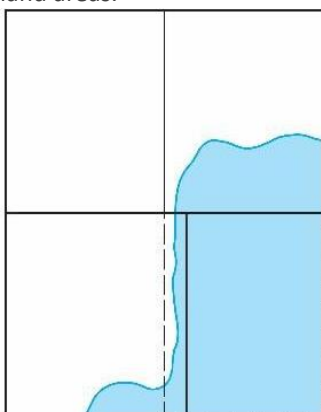


Figure 3: Sheet Line Extension

2.6.4.3 Shifts

A shift is a change in the continuity of sheet lines to accommodate a landmass (Figure 4). Sheets that are shifted usually retain the defined sheet dimensions for the area. A shift may involve more than one sheet. Overlapping sheets are permitted only if the overlapped area does not contain land, as shown in the example on the right.

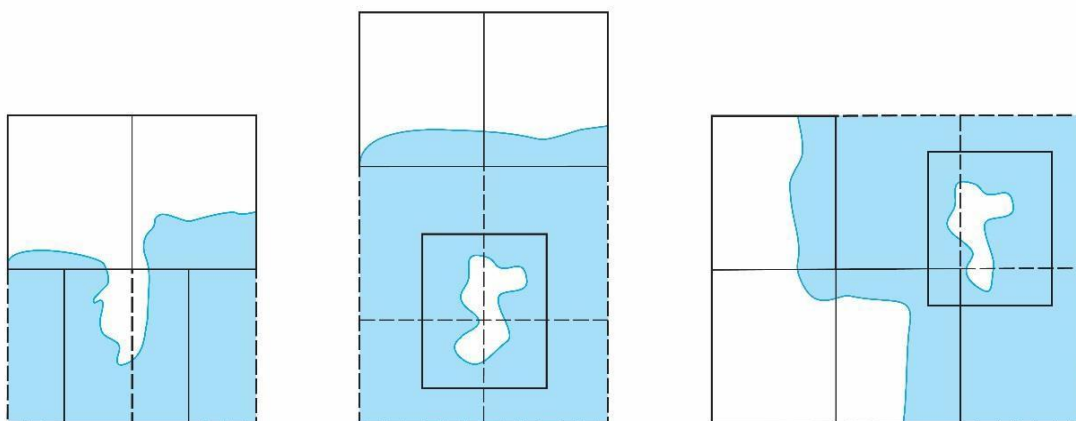


Figure 4: Various Sheet Line Shifts

2.6.4.4 Reproportionment

Reproportionment (Figure 5) permits the adjustment of the latitudinal and longitudinal limits of the defined sheet lines.

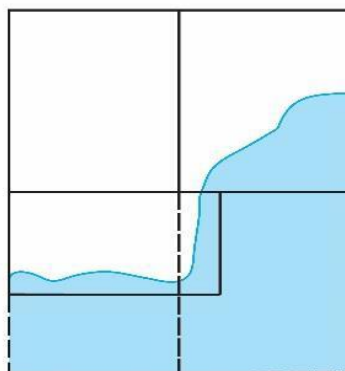


Figure 5: Reproportionment

2.6.4.5 Insets

An inset (Figure 6) is a shift of a portion of a sheet covering an island(s) to relocate it within the open water area of another sheet. Insets should only be shown for consistency with prior editions and when specified in supplemental project instructions. The inset is relocated on the nearest sheet and preferably along the same line of latitude or longitude.

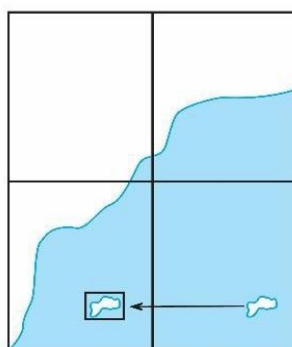


Figure 6: Insets

2.6.5 Labelling

Geographic coordinate labels are shown outside the projection line. The full latitude and longitude values in decimal degree format are shown at the corners of the projection. Fonts, type sizes, and placement guidelines are defined in detail in the style sheets - see - Topographic Map Guidelines Portrayal Catalog (TMS PC) and - Topographic Map Guidelines Annotation Catalog (TMS AC) and Annex D – Topographic Map Guidelines Style Sheets (TMS SS) for examples of this content.

2.7 Data Quality

Components of data quality include currency, consistency, completeness, and accuracy. The KSA topographic map program uses interpreted cartographic point, line, and area features taken from databases maintained or approved by GEOSA. These source databases are multipurpose GIS databases and are not necessarily complete or consistent in the same sense that is represented on historical (paper) topographic map series. The databases are compiled from multiple primary and secondary sources and have no single currency data. KSA topographic maps therefore have different characteristics regarding currency, consistency, and completeness compared to historical (paper) topographic maps. The data tends to be more current than on a legacy map but may be less consistent between maps and between regions because of variations in data collection between different sources.

Accuracy, both positional accuracy and attribute accuracy, of the data is the responsibility of the organization that produced the source data. The overall quality of the cartographic data depends directly on the quality of the source, i.e. GIS databases. KSA topographic maps produced to date do not include a traditional accuracy statement in the map collar because of the variety of data sources used in creating them.

For Cartographic Data Products, GEOSA cartographically adjusts and generalizes the data used in KSA topographic maps where necessary. The XML (Extensible Markup Language) metadata file attached to each map includes data-quality statements in the Data Quality, Accuracy Information, Data Source, and Process Information sections for each data layer.

2.7.1 Accuracy

2.7.1.1 Horizontal accuracy

Horizontal accuracy shall be within 1.0 mm at map scale (for example, 5 meters at 1:5,000) and circular error (CE) shall be at the 90% confidence level.

Circular error is an accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (e.g., horizontal position) will be within the given figure.

2.7.1.2 Vertical accuracy

Vertical accuracy shall be one elevation contour interval linear error (LE) at a 90% confidence level.

Linear error is a one-dimensional error (such as an error in elevation) defined by the normal distribution function.

2.7.1.3 Displaced features

The accuracies stated above shall be for well-defined points such as drainage confluences, major transportation crossing points, survey points, or the equivalent. Feature symbols that are cartographically displaced shall be excluded from the accuracy requirement stated above.

2.8 Data Product Identification

Each topographic map shall be identified primarily by a unique combination of the product name, series identification number, map sheet identification number, and edition number.

Series, sheet, and edition numbers shall be constructed according to the definitions and principles outlined below:

2.8.1 Map series

A series consists of a group of maps which are common to one another in that they:

- Cover a particular geographic area.
- Are on the same sheet line system.
- Are of the same scale or within the same scale group.
- Are prepared under the same cartographic guidelines.

Series is planned to cover all or part of a National, Regional, or Sub-regional area.

Topographic maps are grouped into a map series to facilitate preparation, identification, indexing, storage, and distribution. Each map series shall be identified by a series name and a series number.

2.8.1.1 Series name

The name assigned to a series should normally be the geographic name of the area covered by the series.

Rigid rules cannot be established for the assignment of all series names. With exceptions permissible for necessary deviations, the following guidance applies:

- When more than one series, at the same scale, are designed to cover a region, each series shall be identified by the Regional or local name, qualified by a geographic term. Examples: Tabuk Province, Southern Region, Northern Borders, etc.
- When the series covers a large well-known area, it shall be given the name most commonly used to designate that area. In such cases, the name shall be spelled in accordance with the official name as per the National Geographic Names Database.

When the series covers a small and not widely known area, it shall be identified by the accepted local name.

2.8.1.1.1 Normal sheet naming practices

A map should normally be named after the most outstanding cultural or natural feature within the area of the sheet series name. Names of cultural features are preferred over natural features; however, if a natural feature is better known than any cultural feature appearing on the map, the name of the natural feature should be chosen.

The names of rivers, valleys, mountain ranges, and other features extending over several sheets should be avoided when selecting a sheet name. In addition, area, locality, and tribal names should not be used as a sheet name. Names of destroyed populated places or features should not be used as a sheet name when a natural feature is more prominent.

When the feature is divided by a neatline(s) separating two or more sheets and is the best-known feature on each of the sheets, the feature name is followed by the geographic term (in parenthesis) describing the portion of the feature for which the sheet is being named.

Examples: "MAKKAH (NORTH)" and "MAKKAH (SOUTH)"

2.8.1.1.2 Spelling of sheet name

The sheet name shall be spelled to correspond exactly to the name of the outstanding feature as it is shown in the map interior. Diacritics, hyphens, and apostrophes shall only be shown if they appear with the name on the interior of the map. The name shall be spelled in accordance with the official name as per the National Geographic Names Database.

2.8.1.1.3 Alternate sheet naming practices

When a sheet does not contain a named cultural or natural feature, the name of any adjacent sheet (typically 8 sheets) may be used in conjunction with the appropriate directional term. The adjacent sheet that has the most prominent name shall be selected.

For example: "EAST OF TABUK" shall be used for the sheet immediately to the east of the sheet with the sheet name "TABUK"; "NORTHWEST OF MAKKAH" shall be used for the sheet immediately to the northwest of the sheet with the sheet name "MAKKAH".

In cases where the 8 adjacent sheets are also devoid of any named feature that could be used as a sheet name option, the sheet number of the map being produced may be used as the sheet name.

2.8.1.1.4 Uniqueness of sheet names

Sheet names shall not be duplicated within a map series. Individual maps should be given distinct sheet names wherever possible.

2.8.1.1.5 Coordination of sheet names between scales

Sheet names shall be coordinated whenever possible. For example, in the case of the national map series, the sheet name for the 1:50,000 map should be the sheet name of one of the 1:25,000 maps and the sheet name of the 1:50,000 map should be used as the sheet name for the applicable 1:25,000 map.

2.8.1.1.6 Edition numbering

The edition number shall identify the publication sequence of an individual map. Edition numbers shall run consecutively. A map bearing a higher edition number in the same series is assumed to contain more recent information than the same map bearing a lower edition number.

2.8.1.1.7 Advancement of edition number

The edition number shall be advanced in the following instances:

- Any map on which an alteration or revision is made to the factual data shown on the map or any alteration that affects the operational soundness of the map. Examples would be the addition of a new grid or the revision of boundary information.
- A newly constructed map which is to replace an existing map.
- A map converted from a non-standard scale within the same scale range. For example: a 1:50,000 scale map which replaces a 1:63,360 scale map and retains the same series number.

2.8.1.1.8 Edition numbering for provisional, emergency, special, and temporary printings

The word "edition" shall be used only in conjunction with the edition number. The words "provisional," "emergency," "special," "temporary," etc., shall not be used as prefixes to the word "edition." Such prefixes may be used in conjunction with the word "printing," in which case an edition number is not shown.

2.9 Cartographic Process

The Cartographic Process defines the details of the Cartographic Map Sheet and the definitions and steps required to define the KSA topographic map Sheets and the special cases that may arise.

The following concepts describe a cartographic process for creating a map display (hardcopy or digital) that accurately and effectively portrays feature data and associated metadata (as annotation).

Information on the cartographic process is provided in this TMS to provide a general understanding of the process and to illustrate the terminology used in this TMS. The international standard for the portrayal of geospatial information is ISO 19117, Geographic Information Portrayal ([International Standards Organization, ISO, 2012](#)). That standard indicates that typically there are two types of information included in a geospatial dataset: geographic information (the data) and metadata. Metadata is depicted by annotations.

The design of any map sheet comprising the layout, symbolization, annotation, styling and white space management shall be according to international standards and best practice with adaptation for KSA requirements. For example, the use of English and Arabic text, terminology, special symbols etc. Current cartographic practice, standards and norms within GEOSA shall be incorporated.

2.9.1 Map Collar

The map collar (or map margin) is defined as the area outside the projection line, including all text, diagrams, and other information that appears in this area, except for data related to cartographic coordinate systems, which are logically associated with the projection line and grids.

The general layout of the KSA topographic map collar is defined by the style sheets. Precise content, positions, colors, fonts, and line weights for the KSA topographic map collar information are defined by the style sheets that accompany this guideline - see - Topographic Map Guidelines Portrayal Catalog (TMS PC) and - Topographic Map Guidelines Annotation Catalog (TMS AC) and - Topographic Map Guidelines Style Sheets (TMS SS) for examples of this content.

2.9.2 Map Interior—Data Sources, File Size, and Resolution

All geospatial content on KSA topographic map products is derived from national geospatial databases of GEOSA data programs. Normally, these datasets are owned and hosted by GEOSA, but they may also include data sources owned and hosted by other organizations, provided that these sources have been approved for use by GEOSA.

The KSA topographic map product is freely distributable, but this does not necessarily mean that all the data in the product is in the public domain, as this guideline does not preclude the use of licensed data. All source information and any relevant license or copyright information are documented in the product metadata and listed in the map collar.

Geospatial PDFs produced by the KSA topographic map program must support high-quality printing yet have a manageable file size for storage, viewing, and manipulation in their native digital format. These requirements have to take into account the size of the data sources that are integrated, ground sample distance (GSD), image bit depth, spatial resolution, and file compression.

2.9.3 Map Display

The highest-level concept present in a topographic map is the map display (Figure 7). This section describes the anatomy of a map display, as well as the maps within.

A Map consists of:

- Feature Portrayal (geographic information/feature data),
- Annotation (metadata).

A Map Display consists of 1 or more maps:

- Primary Map (~ map sheet),
- Inset Map(s),
- Margin mini-maps (Annotations).

Each Map in the Map Display has:

- Map Interior (inside neatline),
- Map Margin (outside neatline).

The Map Interior contains:

- Geographic Information (Feature data),
- Annotation (Interior Annotation within the neatline) – see blue outlines:
 - Georeferencing annotations (e.g. grids, graticules),
 - Geo-oriented annotations (e.g. grid labels).

The Map Margin contains:

- Annotation (Interior Annotations related to the neatline),
- Geo-oriented annotations (e.g. grid and graticule ticks and labels),
- Annotation (other),
- Geographic annotations (e.g., "mini-maps", Boundaries diagram, Adjoining Sheets diagram),
- Geo-oriented annotations (e.g., magnetic declination diagram, scale bar, Slope Guide),
- Non-geographic orientations (e.g., symbol legend, copyright notice).

Map Display Design (Style Sheet):

- Specifies content and placement of maps, including the arrangement of interior and other annotations, may employ "open space" over map interiors for the arrangement of inset maps, etc. see Annex D – Topographic Map Guidelines Style Sheet (TMS SS) Examples.

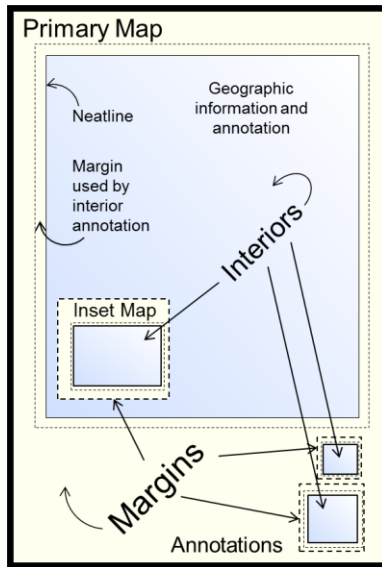


Figure 7: Map Display

2.9.4 Geographic Portrayal Process

All the rules and symbol definitions important to geographic feature portrayal are managed in a portrayal catalog. Whether automated, hand-crafted, or a combination thereof, the process of portraying geospatial information, i.e., attributed geospatial features, can be depicted as shown in Figure 8.

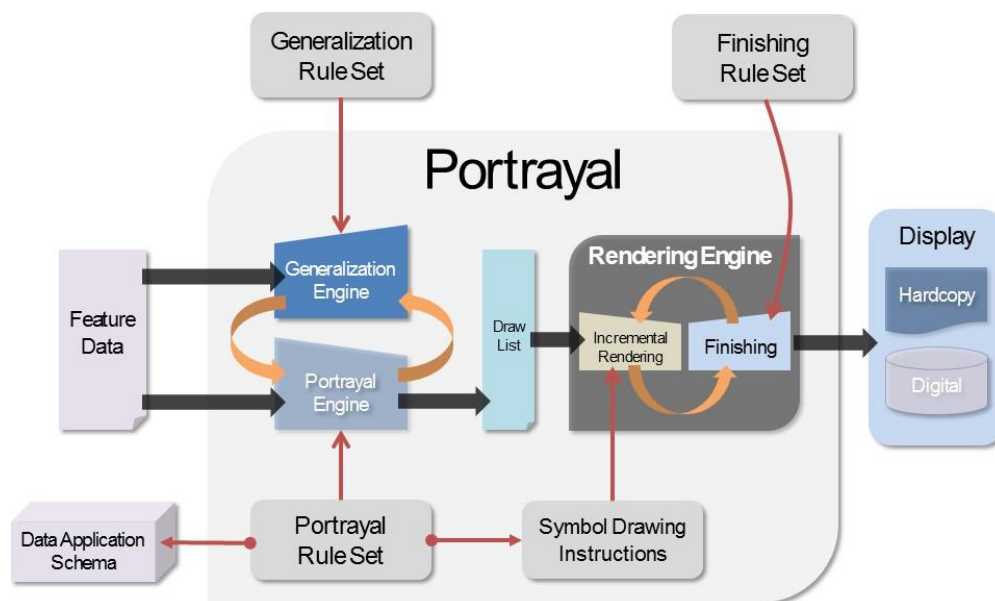


Figure 8: Geographic Portrayal Process as Conceptualized in ISO 19117

Source: ISO 19117

The portrayal of geospatial features and annotations shown on the KSA topographic maps shall be defined in the KSA TMS Portrayal Catalog (TMS PC) and KSA TMS Annotation Catalog (TMS AC). The ISO 19131, Data Product Guideline () allows for multiple portrayal and annotation catalogs to exist, the use of which will result in different portrayals, depending on delivery format and/or user needs.

The ISO 19131 standard also considers a hardcopy map or chart to be a data product.

The Geographic Portrayal Process can be summarized as follows. Feature data that is managed according to a well-defined schema (encoding model) feeds a portrayal process whose first step is to apply generalization rules that guide any changes in the aggregation or delineation of a feature.

Generalization may result in assigning different display geometry to the feature as it would be represented on the portrayal draw list. The portrayal engine coordinates with the generalization engine to determine which portrayal function (portrayal rule) applies to the feature – resulting in the assignment of a symbol code before progressing to the portrayal draw list along with labeling instructions.

Certain Generalization rules may be suited to application of Artificial Intelligence (AI). However such application shall be under the guidance of a skilled cartographer.

The rendering engine takes features from the draw list and applies the associated symbol definitions along with geographic display geometries. To this point, each feature, now represented as a symbol, has been manipulated as an individual entity – with no consideration for other features/symbols on the draw list.

The function of the finishing process is to consider sets of features/symbols that must be made to co-exist in the display space and make adjustments that yield the desired cartographic result. Finishing rules provide a structured way for specifying finishing adjustments. Some symbols must be geospatially displaced to ensure readability when otherwise multiple symbols would “overprint” one another.

Finishing rules will indicate which type of feature stays geospatially accurate and which type of feature gets displaced. Some rules also apply to labeling.

It is the geographic portrayal finishing process that most heavily involves the skills and experience of a cartographer.

Automated methods can achieve various levels of success for the generalization and cartographic displacement of features, but for most demanding projects there is no better solution than the cartographer's judgment and experience. The result of the cartographic process is a map display. A topographic map represents an extensive application of the cartographic process.

The greater map display process encompasses the geographic portrayal process and the annotation process, all of which feed on feature data and metadata as depicted. Certain annotations are georeferenced (e.g., graticule and grids), while certain others are geo-oriented (e.g., declination diagram, road objective). For this reason, the map annotation process includes the geographic portrayal process to help place such annotations in the map display.

2.9.4.1 Series name

The Portrayal Catalog (TMS PC) ([Annex B](#)) presents examples of symbol descriptions, symbol rules, labeling rules, generalization rules, finishing rules, and additional informative guidance associated with feature portrayal on hardcopy topographic maps.

Features portrayed on a TM can be topographic, aeronautical, boundary, elevation, or maritime-related.

2.9.4.2 Annotation Catalog

Very similar to the Portrayal Catalog is an Annotation Catalog. The annotation catalog manages rules for presenting metadata in the map display.

The Annotation Catalog (TMS AC) () presents examples of rules for including, composing, and placing annotations (marginalia) in the map display on hardcopy topographic maps. The TMS AC incorporates information currently found in text and map style sheets. - see - Topographic Map Guidelines Portrayal Catalog (TMS PC) and - Topographic Map Guidelines Annotation Catalog (TMS AC) and Annex D – Style Sheets

2.9.5 Interpreted Cartographic Features

Each KSA topographic map needs to be assembled as a layered geospatial PDF and includes significant features and layers such as transportation, geographic names, elevation, hydrography, boundaries, land cover, and structures. The following list provides examples of features that are included in the KSA topographic maps, though it is a sample and is not intended to be all-inclusive:

- Roads,
- Airports,
- National, Provincial, Municipal, and other Boundaries.
- Contours,
- Coastlines,
- Lakes and ponds,
- Streams and rivers,
- Schools,
- Woodlands,
- Geographic names,
- Land & cadastral boundaries,
- Railroads,
- Trails,
- Hospitals,
- Wetlands.

2.9.6 Orthoimage and Shaded Relief

Each KSA topographic map includes the option to produce an orthoimage layer and a shaded relief layer.

Details of the content of these layers are published separately.

2.9.7 File Size and Resolution

The objective is to produce each KSA topographic map product as a geospatial PDF with a file size of approximately 30 megabytes (Mb). This is an ideal rather than a hard requirement because the data sources and map sheet size will vary across the Kingdom in the case of traditional map series, and will be project dependent in the case of other map sheet products. A file size of 30 MB is small enough for optimal performance and large enough to maintain visual quality.

2.9.8 Layers

The KSA topographic map contains, at a minimum, the following folders and layers:

2.9.8.1 Map Collar

Map Elements (this layer includes all information outside of the map neatline, except for data related to cartographic coordinate systems, including the map title, scale bar, map credit notes, GEOSA logos, and more)

2.9.8.2 Map Frame

Projection and Grids (this layer includes projection coordinate values, geographic and grid ticks, and grid lines).

“Map Collar,” “Map Frame,” and “Images” must be folders and must minimally contain the listed subfolders and layers. The folders and layers listed above are present in every geospatial PDF file and use the precise names given in the preceding list.

The “Map Frame” folder also contains other folders and layers for the cartographic features included in the KSA topographic map, as depicted in Figure 9. These other folders and layers are only shown if the cartographic features are present in a given KSA topographic map.

Compliant with this guideline other data folders and layers may be added as the product develops. Each type of cartographic data in the geospatial PDF file is assigned to a folder or layer. Each folder or layer may be turned on or off by the user.

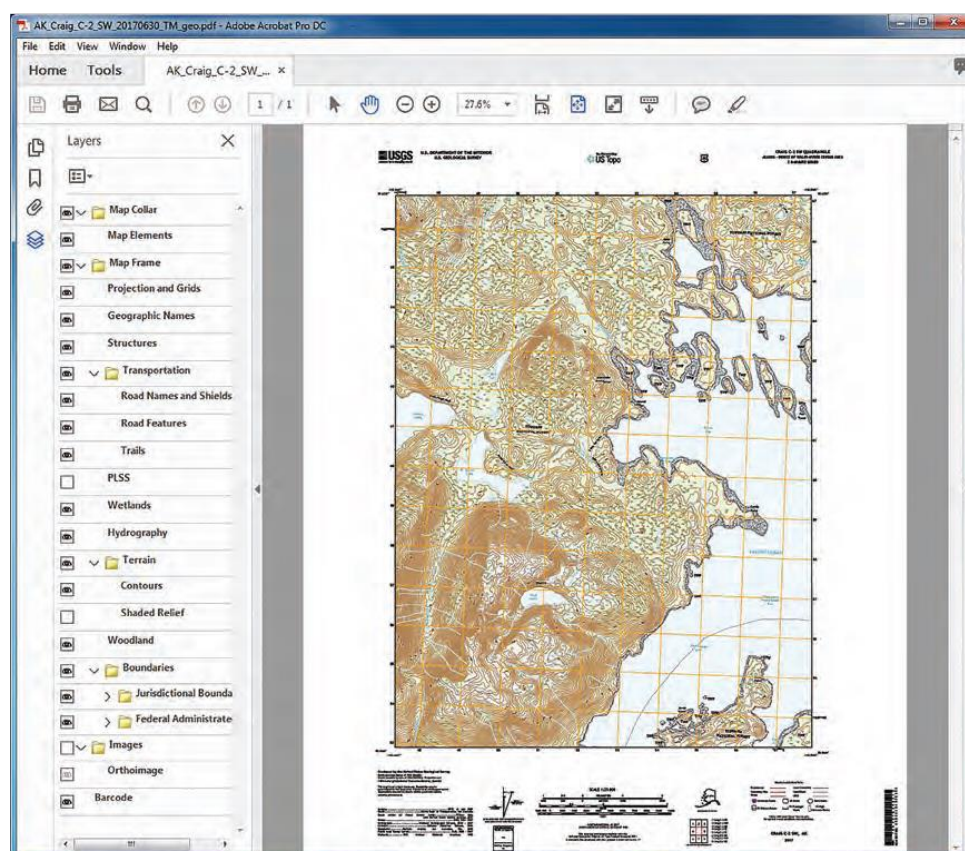


Figure 9: Image of a Topographic Map Depicting Folders and Layers. Images Folder, Shaded Relief Layer

2.9.9 White Space Management

White Space in a cartographic sense refers to the deliberate reservation of space on the map sheet layout that is devoid of text, symbology and other map elements. The purpose is to create visual balance and artistic input to the map sheet design. White Space Management is required to optimise the following:

Visual Contrast & Separation - to visually separate different map components, making it easier for users to distinguish features and read the map. The higher the contrast between features, the more these features will stand out. Conversely, a map that has low visual contrast can be used to promote a more subtle impression. Features that have less contrast will appear to belong together.

Legibility – Legibility refers to the ability of the user to see and understand what is on the map sheet. This requires text and graphical elements, such as lines and symbols, to be easy to find and easy to understand. Visual contrast allows the human eye and mind to find the features and then to distinguish between features, and then the legibility allows for them to be understood.

Balance - to enhance overall aesthetics and visual appearance of the map sheet it is essential that elements are placed within the bounds of the mapsheet to give a visually pleasing balance without crowding or overlapping.

Layer Hierarchy – refers to arranging data layers based on their importance. The order determines which features appear on top or below others. For example, roads and rivers may be prioritized over less significant features like administrative boundaries. Cartographic ordering of features is crucial for effective visualization.

At all times the map sheet shall meet the requirements of legibility, visual contrast, hierarchical organization and balance.

Further reading on these topics can be found at:

- Design principles for cartography – Aileen Buckley - ARGIS Blog, October 2011. <https://www.esri.com/arcgis-blog/products/arcgis-pro/mapping/design-principles-for-cartography/>
- Cartography Chapter 2: Map Elements and Design Principles - Ulirike Ingram 2021 Open Educational Resources for Introduction to Cartographic Processes. Kenneshaw State University.
- Dent, Borden D., Jeffrey S. Torguson, and Thomas H. Hodler. 2009. Cartography: Thematic Map Design, Sixth Edition. Boston, MA: WCB-McGraw Hill, 336 pages, ISBN-13 978-0-07-294382-5, ISBN-10 0-07-294382-3. Pages 207-222.
- Robinson, Arthur H., Joel L. Morrison, Phillip C. Muehrcke, A. Jon Kimerling and Stephen C. Guptill. 1995. Elements of Cartography, Fifth Edition. New York, NY: John Wiley & Sons, Inc., 674 pages, ISBN 0-471-55579-7. Pages 324-330.
- Slocum, Terry, Robert B. McMaster, Fritz C. Kessler, and Hugh H. Howard. 2009. Thematic Cartography and Geographic Visualization, Third Edition. Upper Saddle River, NJ: Pearson Prentice Hall, 561 pages, ISBN-10 0-13-229834-1, ISBN-13 978-0-13-22984-6. Pages 212-221.

2.9.10 Overall Cartographic Process

The overall cartographic process addresses both geographic feature portrayal and annotation to create a complete map display. Figure 10 builds on the geographic feature portrayal process to illustrate this point.

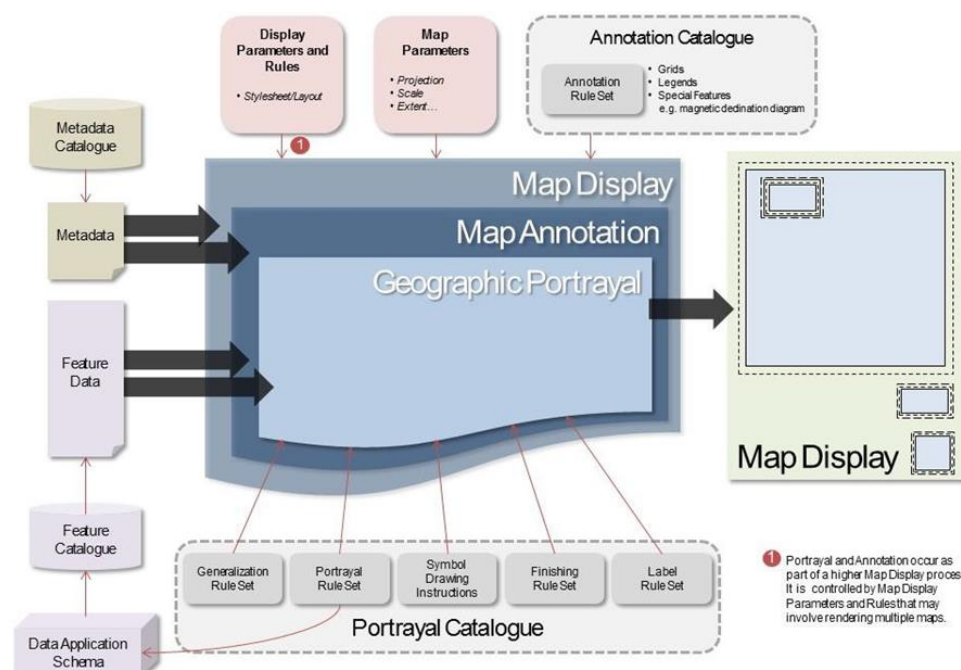


Figure 10: The Cartographic Process for a Complete Map Display

This Guideline is organized in accordance with ISO 19131, Data Product Guideline (). Several of the sections in this Guideline refer to external resources, to document product requirements such as:

- Feature Catalog (TMS FC).
- Topographic Map Portrayal Catalog (TMS PC).
- Topographic Map Annotation Catalog (TMS AC).

Because ISO 19131 allows for multiple portrayals of a TM dataset, the possibility exists to have more than one portrayal catalog, and more than one annotation catalog, for a dataset.

Figure 11 below illustrates the relationship between the components of this TMS.

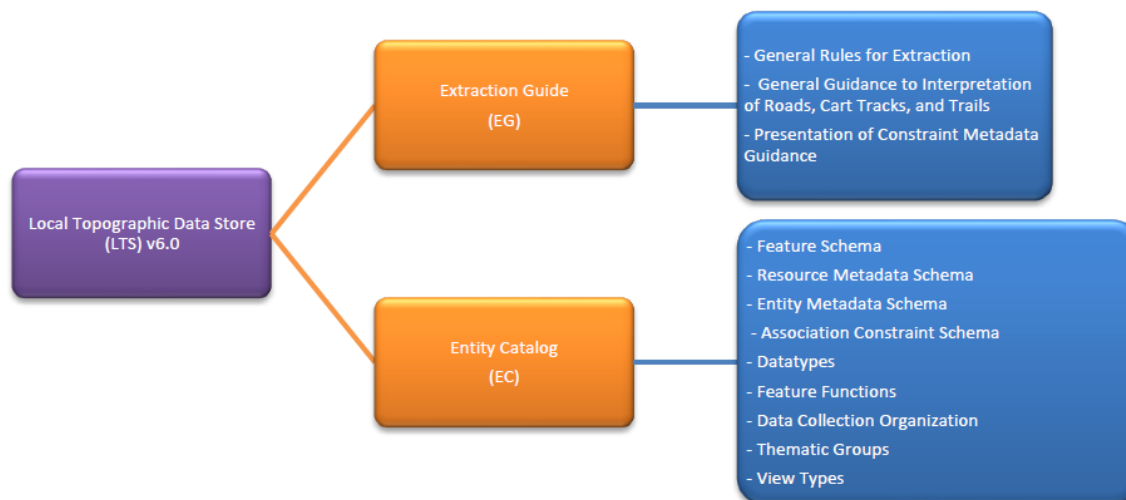


Figure 11: TMS Components

2.10 Data Product Delivery

2.10.1 Delivery medium

KSA topographic maps shall be delivered as digital map sheets in formats such as PDF, and Geospatial PDFs. Alternative formats are Compressed Arc Digitized Raster Graphics (CADRG) or Enhanced Compressed Raster Graphic (ECRG).

KSA topographic maps may also be delivered as printed (or plotted) maps on hardcopy material such as paper or film.

2.10.2 Delivery format

The delivery format shall be a digital version of the map sheet and/or a hardcopy (paper) plotted/printed version of the map sheet.

The KSA topographic maps digital format is a Geospatial PDF. The Geospatial PDF is an open file format, a PDF published by the International Organization for Standardization (ISO) as ISO 32000-2. Geospatial PDF KSA topographic maps can be viewed and printed with Adobe Reader, Adobe Acrobat, or any comparable PDF viewing software.

The product defined by this guideline contains a layered geospatial PDF, an associated metadata file in Extensible Markup Language (XML) format, and a map symbol key (legend) file in PDF. Metadata and symbol key files are bundled with the geospatial PDF as file attachments.

Full-sized examples of style sheets in PDF accompany this guideline in – Topographic Map Guidelines Style Sheets (TMS SS) and provides examples that define the placement of map elements, marginalia, and font sizes and styles for the 1:25,000 scale KSA topographic maps.

The goal of KSA topographic map production is to construct a geospatial PDF that strikes a balance between manageable file size and visual quality.

2.10.3 Work limits

Work limits define the area available for a map to be plotted or printed on a roll or sheet of paper. Dimensions are expressed in linear units of measure.

The normal preferred work limits shall be 558.8 mm (22 inches) east-west and 723.9 mm (28.5 inches) north-south. If necessary, these limits may be increased up to a maximum of 609.6 mm (24 inches) east-west and 749.3 mm (29.5 inches) north-south, to accommodate:

- Wider sheet areas.
- Extended marginalia (i.e., bi/tri-lingual translations); or
- Departures from standard sheet lines.

2.10.4 Paper size and dimensions

Paper/trim size pertains to the overall dimensions on which a map may be plotted and/or cut after printing. Dimensions are expressed in linear units of measure. The dimensions below represent an example of a typical design for producing a map sheet.

- The paper/trim size shall be a total of 12.7 mm (0.5 inches) greater in each direction than the work limits, or 6.35 mm (0.25 inches) around each edge.
- The normal preferred paper/trim size shall be 571.5 mm (22.5 inches) east-west and 736.6 mm (29 inches) north-south.
- If necessary, the maximum paper/trim size may be increased to 622.3 mm (24.5 inches) east-west and 762.0 mm (30 inches) north-south.

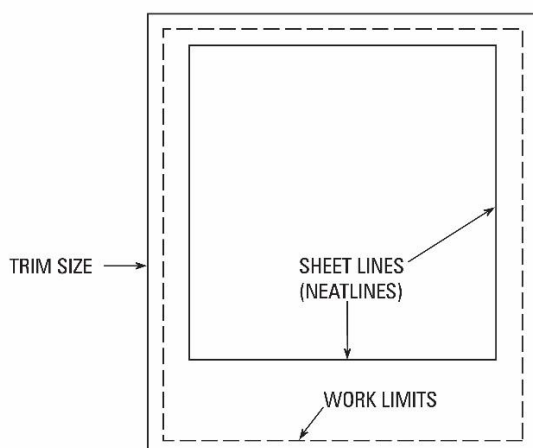


Figure 12: Sheet Lines, Work Limits and Paper/Trim Size

Figure 12 illustrates sheet lines, preferred work limits, and preferred paper/trim size pictorially (see also examples provided in [Annex B - PORTRAYAL CATALOG \(TMS PC\)](#), [Annex D - Topographic Map Specifications Style Sheets \(TMS SS\)](#)).

2.11 Data Maintenance

Data shall be maintained as per the annual map update and maintenance program as determined by GEOSA.

2.12 Metadata

An XML metadata file attached to each geospatial PDF contains standardized language that is determined by the source data used. The content of the XML metadata file is specifically tailored to each map.

The metadata file contains the same information as the map collar, plus additional information as required by the GEOSA metadata content standards. The benefit of duplicating collar information is that the XML file can be parsed by a web-based search engine, making the map and its contents more widely discoverable. (See KSA topographic map collar sections for KSA topographic map collar content in [Section 9 – Cartographic Process](#).)

Metadata file versions are maintained for internal GEOSA use and the content of these files is not defined by this guideline.

KSA topographic map metadata files are named using the following convention:

- SheetName_datestamp_XX_geo.yyy (for example, SheetName_20231205_TM_geo.xml).
- SheetName is a GEOSA defined abbreviation, containing the Province Name and Sheet Name of where the map is located.
- Datestamp is the system-generated date of PDF file creation. Its primary purpose is to make each file name unique, regardless of how many instances of the same product are created for one cell. The timestamp string is a concatenation of year-month-day, where the year has four digits and the other two fields have exactly two digits. No delimiters are used in the timestamp. For example, December 4, 2009, is expressed as 20091204.
- XX is a text string that indicates the map type. Examples are TM (Topographic Map) or OM (Ortho Map). The full domain of values for this string is to be specified by GEOSA.
- Geo is a text string that indicates the PDF is georeferenced.
- YYY is the filename extension associated with the file type. A text string of “XML” indicates an Extensible Markup Language (XML) file, and a text string of “pdf” indicates a Portable Document Format (PDF).

No requirements are specified for letter case; file names may use any combination of uppercase and lowercase letters.

2.13 Security Considerations

2.13.1 Classification

The degree of classification is determined in accordance with the provisions of the Ministry of Defense. The appropriate classification shall be indicated in the security classification guidance for the project.

The security classification of the products generated using this guideline should be the lowest category practicable – normally UNCLASSIFIED with some form of restricted dissemination.

2.13.2 Classification Policy

Each map bearing a security classification marking shall also identify the classifier and shall also contain downgrading or declassification instructions. The appropriate note or statement shall be determined in accordance with the provisions of the Ministry of Defense Regulations. The specific note shall be indicated in pertinent security classification guidance for the project.

2.13.3 Handling Description

Certain maps, classified or unclassified, require notes which restrict their distribution. When required, the appropriate note shall be specified in the security classification guidance about the project.

A Restricted Dissemination or LIMITED DISTRIBUTION note is sometimes required on unclassified maps. The exact wording of this note is subject to change. Refer to current guidance and/or supplementary project instructions for the text of this note.

The point of contact for Topographic Map data products shall be contained in a “users note”. See TMS Annotation Catalog (TMS AC) for more details.

Since the point of contact information contained in the user's note changes frequently, the Point of Contact for this data product guideline should be contacted to obtain the latest information.

2.14 Additional information

This section provides Additional Information as needed.

3. Supplementary Informative Elements



3.1 Annex A - Topographic Map Guideline Feature Catalog (TMS FC) Examples

This Annex A provides Topographic Map Guidelines Feature Catalog (TMS FC) examples to accompany Annex B - Topographic Map Guidelines Portrayal Catalog (TMS PC) examples, Annex C - Topographic Map Guidelines Annotation Catalog (TMS AC) examples and Annex D – Style Sheet examples of the KSA Guidelines for Topographic Maps. This Feature Catalog contains a subset of descriptive information for each feature contained in the TMS Guideline.

This Annex provides a framework guideline to serve as a container for Standard Operating Procedures (SOP's) for cartographic maps as and when these are created.

Examples of Feature Types for the 1:25 000 and 1:50 000 Scale Topographic Maps are provided.

The content is in 2 parts:

- Examples of feature name and feature code, followed by a definition and description (when applicable) of that feature. Following the feature information is a listing of all of the attributes used in the symbol, labeling, finishing and/or generalization rules for the respective feature.
- Examples of Attribute Definition and Description - lists all of the attributes of the feature name & feature code, followed by the definition and description of that attribute. A <DESC> identifier (when applicable) is used to denote when the attribute's definition end and the attribute's description begins.

The content and examples are adopted from the National Geospatial Intelligence Agency (NGA) Standards.

3.1.1 Feature Catalog Examples

3.1.1.1 Examples of Feature Name and Feature Code

Examples of Feature Name and Feature Code, followed by a definition and description (when applicable) of that feature. Following the feature information is a listing of all of the attributes used in the symbol, labelling, finishing and/or generalization rules for the respective feature.

Administrative Boundary (FA000)

Definition: A boundary between administratively controlled regions.

Attributes: Boundary Status [BST], Geopolitical Entity : Geographic Name Information (1) : Full Name (first side) [ZI005_FNAA], Geopolitical Entity : Geographic Name Information (2) : Full Name (second side) [ZI005_FNAB], Geopolitical Line Type [LSP]

Administrative Subdivision (FA003)

Definition: An administratively subordinate subdivision of a geopolitical entity.

Description: A geopolitical entity (country) is typically divided into first-, second-, and lower-order administrative subdivisions. First-order administrative subdivisions are immediately subordinate to the government of the geopolitical entity, with second- and lower-order divisions subordinate to those above them. Examples: (first-order) a United States state, a German Land, a French region; a Canadian province; (second-order) a U.S. county, a French department; (third-order) a U.S. township, a French arrondissement; (lower-levels) a French commune.

Attributes: BGN Administrative Level [BAL], Geographic Name Information : Full Name [ZI005_FNA]

Aerial Farm (AT012)

Definition: A collection of aerals that are collocated and serve a common purpose.

Description: They may be organized either to function as a single larger virtual device (for example: a phased array) or function relatively independently of each other (for example: pointed in different directions and operating at different frequencies as at a satellite communication ground station).

Attributes: Angle of Orientation [AOO], Area [ARA], Physical Condition [PCF]

Aerodrome Beacon (GB013)

Definition: A beacon used to indicate the location of an aerodrome from the air.

Description: Aerodrome beacons may consist of either a rotating light source or a strobe light.

Attributes: Navigation Light Characteristic [CHA]

Figure 13: Examples of Feature Name and Feature Code

3.1.1.2 Examples of Attribute Definition and Description

Examples of Attribute Definition and Description – Attribute Name followed by the Definition and description of that attribute. A <DESC> identifier (when applicable) is used to denote when the attribute’s definition ends and the attribute’s description begins.

Attribute	Definition and/or Description
Aerodrome Pavement Information : Aerodrome Movement Area Surface Composition [ZI019_ASU]	The type of the predominant material of which a surface of the movement area is composed. [DESC] Example surfaces include: asphalt, concrete, or grass.
Airfield Symbol Type [ASY]	The type of airfield symbol designation.
Amusement Attraction Type [AMA]	The type of an amusement park attraction based on its geometric form, appearance, configuration, and/or use.
Angle of Orientation [AOO]	The angular distance in the horizontal plane measured from true north (0 degrees) clockwise to the major axis of the feature. [DESC] If the feature is square, the axis 0 up to 90 degrees is recorded. If the feature is circular, 360 degrees is recorded.
Aqueduct Type [ATC]	The type of an aqueduct based on its structure.
Area [ARA]	The area within the delineation of the feature.
Base Elevation [BEL]	The vertical distance from a specified vertical datum to the terrain surface of or at the base of the feature. [DESC] If the feature is not supported above the surface by another feature then the base of the feature is usually located at ground or water level on the downhill/downstream side. For non-inland water bodies, the water level is usually understood to be Mean Sea Level (MSL). In the case of a survey marker (monument) this is usually the elevation assigned to the marker (monument).

Figure 14: Examples of Examples of Attribute Definition and Description

3.2 Annex B - Topographic Map Guidelines Portrayal Catalog (TMS PC) Examples

This Annex provides the structure (with examples) required for a Topographic Map Guidelines Portrayal Catalog (TMS PC).

3.2.1 Overview

This Topographic Map Guidelines Portrayal Catalog (TMS PC) defines requirements for the portrayal of Feature Data on a Topographic Map and serves to provide guidelines for the Portrayal Rules to describe the content and arrangement. The document provides structure and examples of the content required to define Portrayal.

This Annex provides a framework guideline, structure of the sections and content required for establishing a Topographic Map Guidelines Portrayal Catalog (TMS PC). This Annex serves as a container for Standard Operating Procedures (SOP's) as and when these are created. Typical headings are listed and examples for 1:25,000 and 1:50,000 scale provided.

The content and examples are adopted from the National Geospatial Intelligence Agency (NGA) Standards.

It is a non-normative presentation of information and provides additional detailed descriptions of data elements. See Annex A -Topographic Map Guideline Feature Catalog (TMS FC) Examples for a listing of the features to be portrayed on a TM. Features not listed shall not be portrayed.

For the full detail of the NGA Guideline refer to The NGA STANDARDIZATION DOCUMENT - DATA PRODUCT SPECIFICATION (DPS) - 1:25,000; 1:50,000 and 1:100,000 Scale Topographic Map (TM) – Portrayal Catalog (PC) (2021-07-31) Version 1.2 (NGA.STND.0035-2_1.2_DPSTMPC).

The various features shown on the map are represented by standard NGA.STND.0035-2_1.2_DPSTMPC symbols. These symbols are explained in the margin of the map along with other information about the map and its content. Topographic map content includes symbolization of transportation and cultural features, vertical obstructions, hydrography, hypsography, vegetation, boundaries, geographic place-names.

3.2.2 Feature-specific Portrayal Guidance

This provides additional clarification guidance to feature portrayal topics not otherwise contained in either Feature Symbols or Finishing, Generalization and Labeling Rules.

Clarification for Transportation, (Roads, Bridges, Overpasses and Tunnels), Railways, Culture, Hydrography, Hypsography/Physiography, Boundaries.

3.2.3 Geographic Names and Labelling

This section provides guidance for the treatment of geographic names, descriptive information, and expressions of political status on maps at the standard scales of military mapping. Geographic names related data includes the identification of geographic features portrayed on the map, descriptive terms, administrative division and sovereignty nomenclature, and statements of political and administrative status, as well as certain information that appears in the map margin.

It includes the source & importance of geographic names, policies for application of geographic names, labelling, principles of label placement, labeling/naming of populated places, labeling of point and small surface features, labeling of linear (curve) features, labeling of survey points, spot & elevation contours, labeling of hypsographic features, labeling of vegetation features, labeling of enclosures, area/locality & tribal names, labeling of water bodies, marshes/bogs, and swamps, labeling of coastal landforms, road route markers, UNESCO world heritage sites.

3.2.4 Portrayal Catalog

The Portrayal Guideline should also include details in separate sections for:

- **Feature Symbols:** provides feature portrayal information, sorted alphabetically by feature name. This provides users with an overall view of example symbols for each feature type and shall serve as an index to assist users in locating the detailed guidelines for Feature Symbols.
- **Color:** defines the colors used in the TMS PC in several well-known color guideline methodologies.
- **Text Characteristics:** defines the characteristics of the text used for labels on Feature Symbols.
- **Area Components:** lists each of the area components used in Feature Symbols.
- **Line Components:** lists each of the line components used in Feature Symbols.
- **Point Components:** lists each of the point components used in Feature Symbols.
- **Detailed Drawing Instructions:** provides the detailed guidelines for reusable graphic icons.

3.2.5 Examples:

3.2.5.1 Feature-specific Portrayal Guidance Example

2.2 Culture

2.2.1 Important buildings

Buildings that are important because of their military significance, cultural importance, or unique appearance and/or construction can be identified by a unique symbol and/or labeling.

2.2.1.1 Important buildings in Built-Up Areas

Where numerous important buildings have been extracted in a Built-Up Area and their portrayal will produce a situation of map illegibility, only the most landmark or prominent buildings shall be retained. It is undesirable to have a profusion of important buildings, especially ones that require labeling, in these areas. Where a selection is required, those that are visible from afar should have first preference for retention. This would include the tallest, the largest and/or the most uniquely shaped building within the area of interest.

Figure 15: Feature-specific Portrayal Guidance - Important Buildings Example

3.2.5.2 Symbol Description Example

Road / intact / divided / narrow separation (curve)



Symbol Graphic

Symbol Placement Instructions: The symbol is applied to a line equidistant between the two extracted traveled ways of the divided road, if two roads are extracted. If a single road is extracted, the symbol is applied to the geometry of the feature.

Symbol Type: Compound

Note: The symbol is a combination of four lines. Display Level from bottom to top: Component L00429, Component L00432, Component L00010, Component L00434. No symbol is portrayed if the Vertical Relative Location [LOC] = Completely Below Ground Surface [40].
*Centerline Spacing [MWG] <=30 meters at 1:25,000 scale, <= 60 meters at 1:50,000 scale, and <= 120 at 1:100,000 scale.

Component Ref #:	L00429	Component Type:	Line - Simple	Color Token:	Black
Description:	Black 1.25mm Solid Line				
Parameters:					
Line Weight:	1.25mm				
Component Ref #:	L00432	Component Type:	Line - Simple	Color Token:	Dk-Brown202
Description:	Dk-Brown202 0.95mm Solid Line				
Parameters:					
Line Weight:	0.95mm				
Component Ref #:	L00010	Component Type:	Line - Simple	Color Token:	Black
Description:	Black 0.25mm Solid Line				
Parameters:					
Line Weight:	0.25mm				
Component Ref #:	L00434	Component Type:	Line - Simple	Color Token:	PaperWhite
Description:	PaperWhite 0.15mm Solid Line				
Parameters:					
Line Weight:	0.15mm				

Figure 16: Symbol Description Example

3.2.5.3 Symbol Rule Example

Symbol Rules		
Reference #:	Description:	Criteria:
PS-00131	Bridge / non-fixed / non-pedestrian (curve)	Bridge Opening Type [BOT] = (Bascule [4], Drawbridge [15], Lift-bridge [11], Opening [16], Retractable [12], Swing-bridge [10])
	AND	Transportation System Type [TRS] = (Aqueduct [2], Canal [5], No Information [-999999], Other [999], Pipeline [10], Railway [12], Road [13], Runway [25], Taxiway [20])

Figure 17: Symbol Rule Example

3.2.5.4 Label Rules Example

Label Rules		
Label Placement Rules (see Annex J): LP-3506, LP-4813, LP-9999		
Reference #:	Description:	Criteria:
PL-00627_1	Space Facility / intact (surface)	Physical Condition [PCF] = (Intact [2], No Information [-999999])
	Label String: [Geographic Name Information : Full Name] "Space facility"	
	Characteristics: 1, 1	
	1 (Black, Condensed, 6 pt, Mixed case)	
PL-00627_2	Space Facility / not intact (surface)	Physical Condition [PCF] = (Construction [1], Damaged [4], Destroyed [6], Dismantled [5], Unmaintained [3])
	Label String: [Geographic Name Information : Full Name] "Space facility" "(" [Physical Condition] ")"	
	Characteristics: 1, 1, 22, 22, 22	
	1 (Black, Condensed, 6 pt, Mixed case)	
	22 (Black, Condensed, 6 pt, Lower case)	

Figure 18: Label Rules Example

3.2.5.5 Symbol Examples

Forest / deciduous (surface)



Symbol Graphic

Symbol Placement Instructions: The symbol is applied to the geometry of the feature.
 Symbol Type: Compound

A

Note: Components A00009 and A00011 are combined into one symbol. Display level from bottom to top: Component A00009; Component A00011.

Component Ref #:	A00009	Component Type:	Area - Fill	Color Token:	Green362-31
Description:	Green362-31 area Fill				
Parameters:					
Screen Angle:	45 degrees				
Screen Percent:	31%				
Component Ref #:	A00011	Component Type:	Area - Pattern	Color Token:	Green362
Description:	Green362 Deciduous Tree Fill				
Parameters:					
Area Pattern:	AP-54, see Annex I				

Symbol Rules

Reference #:	Description:	Criteria:
PS-00338	Forest / deciduous (surface)	Canopy Cover [DMT] >= 51
	AND	Foliage Type [TRE] = Deciduous [1]

Bridge Tower (AQ055)

Bridge Tower / obstruction (point)



Symbol Graphic

Symbol Placement Instructions: Origin: Center of solid circle
 Symbol Type: Simple

B

Note: If the symbol is moved away from the associated solid circle element, a Leader Line / blue072 is added connecting the symbol and the solid circle.

Component Ref #:	P00008	Component Type:	Point - Shape	Color Token:	Blue072
Description:	Vertical obstruction				
Parameters:					
Note:	See Annex H for detailed drawing instructions.				

Symbol Rules

Reference #:	Description:	Criteria:
PS-00089	Bridge Tower / obstruction (point)	Height Above Surface Level [HGT] >= 46 meters (150 feet)

Label Rules

Label Placement Rules (see Annex J): [LP-9992](#)

Reference #:	Description:	Criteria:
PL-00089_1	Bridge Tower / obstruction (point)	N/A
	Label String:	"Bridge tower" [Highest Elevation] "([Height Above Surface Level])"
	Characteristics:	15 46 95, 95, 95 15 (Blue072, Condensed, 6 pt, Mixed case) 46 (Blue072, Condensed, Bold, 6 pt, Lower case) 95 (Blue072, Condensed, 6 pt, Lower case)

Finishing Rules

Finishing Rules (see Annex J): [FL-9986](#), [FL-9992](#), [FR-0046](#)

Symbol Samples



Figure 19: Symbol Examples for Compound (21A) & Simple Symbol (21B) Type.

3.2.5.6 Symbol Graphic Code Example

Symbol	Number	Feature Code	Sample Graphic
Elevation Contour / land / mound / intermediate (curve)	PS-00359	CA010	
Elevation Contour / land / depression / index (curve)	PS-00354	CA010	

Figure 20: Symbol Graphic Code Example

3.2.5.7 Color Token Codes Example

Color Token Name	Color Description	Pantone Ink Code	Red-Light	Red & Blue / Green-Light	C	M	Y	K	HTML	R	G	B	SPC Ink Code
Black	Black	Process Black U	☑	☑	0	0	0	100	#4B4B4B	75	75	75	58600
Black-07	Black 7%	Process Black U	☑	☑	0	0	0	7	#E0E0E0	224	224	224	58600
Black-12	Black 12%	Process Black U	☑	☑	0	0	0	12	#D3D3D3	211	211	211	58600
Black-21	Black 21%	Process Black U	☑	☑	0	0	0	21	#C2C2C2	194	194	194	58600
Black-42	Black 42%	Process Black U	☑	☑	0	0	0	42	#9E9E9E	158	158	158	58600
Black-54	Black 54%	Process Black U	☑	☑	0	0	0	54	#8B8B8B	139	139	139	58600
Blue072	Blue	Blue 072 U	☑	☑	92	70	0	0	#38469D	56	70	157	46351
Blue072-12	Blue 12%	Blue 072 U	☑	☑	11	8	0	0	#C0C7E9	192	199	233	46351

Figure 21: Color Token Codes Example

3.2.5.8 Line Components Example

Line - Styled			
L00029			
Description: Black 0.1mm Dashed Line (0.1mm wt, 2.0mm dash, 0.5mm gap)			Color Token: Black
Line Weight: 0.1mm	Dash Length: 2.0mm	Dash Spacing: 0.5mm	
L00030			
Description: Black 0.2mm Dashed Line (0.2mm wt, 1.5mm dash, 0.5mm gap)			Color Token: Black
Line Weight: 0.2mm	Dash Length: 1.5mm	Dash Spacing: 0.5mm	
L00031			
Description: Black 0.2mm Dashed Line (0.2mm wt, 1.0mm dash, 0.5mm gap)			Color Token: Black
Line Weight: 0.2mm	Dash Length: 1.0mm	Dash Spacing: 0.5mm	
L00032			
Description: Black 0.2mm Dashed Line (0.2mm wt, 2.0mm dash, 0.5mm gap)			Color Token: Black
Line Weight: 0.2mm	Dash Length: 2.0mm	Dash Spacing: 0.5mm	
L00035			
Description: Black 0.25mm Dashed Line (0.25mm wt, 3.5mm dash, 0.5mm gap)			Color Token: Black
Line Weight: 0.25mm	Dash Length: 3.5mm	Dash Spacing: 0.5mm	

Figure 22: Line Components Example

3.2.5.9 Detailed Drawing Instruction Example

Description: **Administrative Boundary / international**
Component Ref #: L00122, L00123 Color Token: Black, Dk-Brown202-54
Origin: Center of dashed line
Overall component width: 2.00mm
Length of stem or segment: 2.00mm, 7.00mm
Lineweight of stem or base: 2.00mm
Lineweight (all or other): 0.50mm
Space between elements: 1.00mm

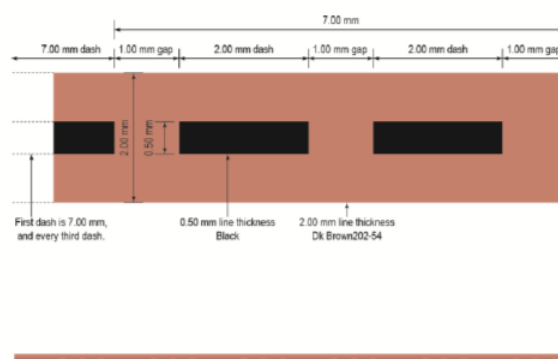


Figure 23: Detailed Drawing Instruction Example

3.2.5.10 Area Pattern Example

Description: **Mangrove**
Component Ref: A00025
Color Token: Green346
Tile Size: 20.0mm x 20.0mm
No. of Components: 11
MIL-STD-2410 Reference: AP-8
Note: Pattern shown with A00007.

Lineweight: 0.10mm

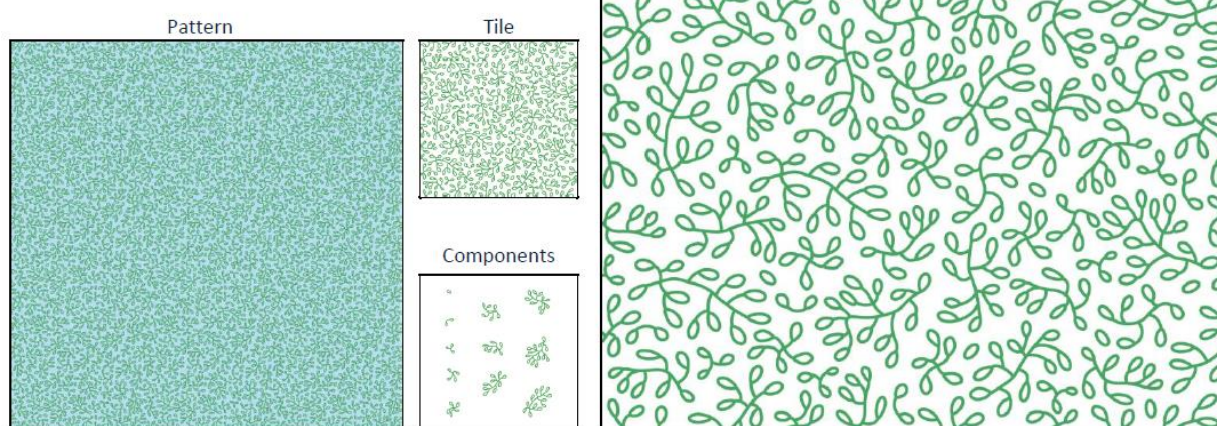


Figure 24: Area Pattern Example

3.2.5.11 Finishing Displacement Rules Example

FD - Finishing Displacement rules

- FD-1510** When a Road (AP030), Cart Track (AP010) or Trail (AP050) enters a "hairpin turn" condition, such as in a steep mountainous region, the coalescing symbol shall be displaced 0.15 mm apart (symbol - edge to edge).
- FD-1650** If two or more Railways (AN010) or Railway Sidetracks (AN050) are on separate roadbeds, and the symbols coalesce, the spacing between symbols shall be increased to 1.0 mm. If the Railway and Railway Sidetrack coalesce, the Railway Sidetrack shall be displaced to 1.0 mm from the Railway. If necessary, the placement of the repeating ticks on the Railway and/or Railway Sidetrack shall be adjusted (offset) along the linear symbol to avoid overprinting.
- FD-9995** When a Culvert (AQ065) crosses beneath a line/curve Aqueduct (BH010), Cart Track (AP010), Ditch (BH030), Ice Route (AQ075), Moat (BH100), Navigable Canal (BH020), Penstock (BH110), Pipeline (AQ113), Railway (AN010), Railway Sidetrack (AN050), Trail (AP050), or Water Race (BH065), the apex of the "V" symbol shall be displaced 0.2 mm from the feature symbol.

Figure 25: Finishing Displacement Rules Example

3.2.5.12 Finishing Label Rules Example

FL - Finishing Label rules

- FL-0062** Area/surface Aqueduct, (BH010), Ditch (BH030), Navigable Canal (BH020), and River (BH140), shall be labeled with upper case type centered within its limits, with type size proportional to the width (WID) of the feature. If the feature is too narrow to accommodate the minimum size type, then the type shall be placed 0.5 mm above (or below, if necessary) and parallel to the feature. Repeat label approximately every 150 mm to 300 mm for either situation, or at least two times, length of feature and label permitting. In either condition (label within or above/below the feature), align the type when necessary to the curvature of the feature. If the feature changes back and forth between an area/surface and a line/curve geometry, the type style shall change from upper case type for the area/surface portions, to upper and lower case type for the line/curve portions. The repeat dimensions remain the same.

Figure 26: Finishing Label Rules Example

3.2.5.13 Finishing Representation Rules Example

FR - Finishing Representation rules

- FR-0005** If Angle of Orientation (AOO) is not available (or is No Information, -999999), point Building (AL013) symbols and other building-like point symbols shall be oriented perpendicular (90 degrees) to all line/curve and area/surface symbols if any portion of the point symbol is < 0.2 mm from the line/curve or area/surface symbol, EXCEPT for the following features: Administrative Boundary (FA000), Administrative Subdivision (FA003), Anchorage (BB010), Annotated Location (ZD045), Aquatic Vegetation (BD061), Brush (EB070), Cable (AT005), Cableway (AT041), Cane (EC010), Cleared Way (EC040), Conservation Area (FA210), Cropland (EA010), Elevation Contour (CA010), Ferry Crossing (AQ070), Foreshore (BA023), Forest (EC015), Forest Clearing (EC060), Geopolitical Entity (FA002), Grassland (EB010), Hop Field (EA055), Ice Shelf (BJ065), Industrial Farm (AL270), Moraine (MJ020), Named Location (ZD040), Oasis (EC020), Orchard (EA040), Railway Yard (AN060), Rapids (BH120), Reef (BD120), Rice Field (BH135), Rock Formation (DB160), Sand Dunes (DB170), Soil Surface Region (DA010), Steep Terrain Face (DB010), Thicket (EB020), Tundra (BJ110), Vineyard (EA050), Void Collection Area (ZD020), and

Figure 27: Finishing Representation Rules Example

3.2.5.14 Finishing Suppression Rules Example

FS - Finishing Suppression rules

- FS-0103** When a Railway (AN010) or Railway Sidetrack (AN050) and a Road (AP030) coincide or coalesce at map scale when on the same Bridge (AQ040), the Railway or Railway Sidetrack shall be suppressed to a distance of 0.25 mm back from the wing ticks at each end of the bridge.
- FS-2024** The Built-up Area (AL020) and Settlement (AL105) outline shall be suppressed if coincident with Aqueduct (BH010), Bog (BH015), Bridge (AQ040), Cart Track (AP010), Dam (BI020), Ditch (BH030), Fence (AL070), Glacier (BJ030), Inland Waterbody (BH082), Land Subject to Inundation/controlled (BH090), Land Water Boundary (BA010), Mangrove Swamp (ED030), Lock (BI030), Marsh (ED010), Navigable Canal (BH020), Railway (AN010), Railway Sidetrack (AN050), River (BH140), Road (AP030), Sabkha (BH160), Salt Evaporator (BH155), Salt Flat (BH150), Shoreline Construction (BB081), Shoreline Ramp (BB082), Snow Ice Field (BJ100), Sports Ground (AK040), Swamp (ED020), Trail (AP050) or Wall (AL260) line/curve symbol or outline.

Figure 28: Finishing Suppression Rules Example

3.2.5.15 Finishing Thinning Rules Example

FT - Finishing Thinning rules

- [FT-0300](#) If less than four individual point symbols of the same feature coalesce and they cannot be displaced (see [FD-9999](#)) due to the proximity of other feature symbols, only one symbol shall be shown. If four or more individual point symbols of the same feature coalesce, a representative pattern of the symbols shall be shown. The point symbols located at the furthest or greatest extent of the coalescing symbols shall be retained when thinning (i.e. retain buildings at the start and end of a thinned line of buildings).
- [FT-9995](#) Multiple parallel Pipeline (AQ113) symbols shall be thinned per [FT-9996](#). To avoid incorrect or illogical portrayal, the Vertical Relative Location [LOC], Physical Condition [PCF], and Physical Product [PPO] of the respective pipelines, when known, shall be taken into consideration when representing multiple pipelines (i.e. combining water and petroleum pipelines). When thinning pipeline symbols, the priority for retention shall be: 1. above surface; 2. on surface; 3. below surface.

Figure 29: Finishing Thinning Rules Example

3.2.5.16 Generalized Change Rules Example

GC - Generalization Change rules

- GC-9978** For area/surface Buildings (AL013) with a Feature Function (FFN) of Checkpoint, Health Office, Hospital, Railway Passenger Transport, or Railway Transport; or a Religious Facility Type (ZI037_RFA) of Minaret, Pagoda, Stupa, or Temple; and if the area (ARA) is less than the square meters indicated for the scale (1:25,000 – 1,405 sq. m; 1:50,000 – 5,625 sq. m; 1:100,000 – 22,500 sq. m), the area/surface delineation shall be replaced with a point delineation. The point delineation shall be placed at the centroid of the area/surface feature.
- GC-9979** Replace the area/surface delineation with a line/curve delineation. The line/curve delineation shall be placed equidistant from each side of the length of the area/surface feature.
- GC-9981** If the length (LZN) is less than the number of meters indicated for the scale (1:25,000 – 25 m; 1:50,000 – 50 m; 1:100,000 – 100 m), then the length shall be increased to that number of meters. If the width (WID) is less than the indicated number of meters for the scale (1:25,000 – 20 m; 1:50,000 – 40 m; 1:100,000 – 80 m), then the width shall be increased to that number of meters. Lengths and widths shall be increased equally in each direction from the centroid of the area/surface feature.

Figure 30: Generalized Change Rules Example

3.2.5.17 Label Placement Rules Example

LP - Label Placement rules

- LP-0050** Variable type sizes shall be placed for area/surface features per area/surface sizes at map/chart scale: 6 point $\leq$ 770 mm sq. area and $\leq$ 14 mm width; 7 point $\leq$ 2,300 mm sq. area and $\leq$ 28 mm width; 8 point $\leq$ 3,600 mm sq. area and $\leq$ 36 mm width; 9 point $\leq$ 5,200 mm sq. area and $\leq$ 44 mm width; 10 point $\leq$ 9,800 mm sq. area and $\leq$ 62 mm width; 12 point $\leq$ 16,650 mm sq. area and $\leq$ 84 mm width; 14 point $\leq$ 25,000 mm sq. area and $\leq$ 105 mm width; 16 point $\leq$ 36,000 mm sq. area and $\leq$ 125 mm width; 18 point $>$ 36,000 mm sq. area and $>$ 125 mm width. The type size may be adjusted depending on the length of the character string. However in no case shall the type size be less than 6 point.
- LP-0051** Variable type sizes shall be used for line/curve features per line/curve length at map/chart scale: 6 point - $\leq$ 80 mm length; 7 point - 81-120 mm length; 8 point 121-160 mm length; 9 point - $>$ 160 mm length. The type size may be adjusted depending on the length of the character string. However in no case shall the type size be less than 6 point.
- LP-0062** Area/surface Aqueduct, (BH010), Ditch (BH030), Navigable Canal (BH020), and River (BH140), shall be labeled with upper case type centered within its limits, with type size proportional to the width (WID) of the feature.

Figure 31: Label Placement Rules Example

3.2.5.18 Feature Name / Alternate Name Listing Example

NFDD Feature Name / Alternate Name	NFDD Feature Name	Feature Code	Page
Administrative Boundary	Administrative Boundary	FA000	A-1
Administrative Division	Administrative Subdivision	FA003	A-4
Administrative Subdivision	Administrative Subdivision	FA003	A-4
Aerial	Tower	AL241	A-455
Aerial Cableway	Cableway	AT041	A-82
Aerial Farm	Aerial Farm	AT012	A-6
Aerial, Dish	Dish Aerial	AT010	A-125
Aerodrome Beacon	Aerodrome Beacon	GB013	A-8
Aerodrome, Land	Land Aerodrome	GB005	A-247

Figure 32: Feature Name / Alternate Name Listing Example

3.2.5.19 Feature Code Listing Example

Feature Code	NFDD Feature Name	Page
AM040	Mineral Pile	A-268
AM060	Surface Bunker	A-439
AM065	Munition Storage Facility	A-277
AM070	Storage Tank	A-434
AM071	Tank Farm	A-446
AM075	Fuel Storage Facility	A-205

Figure 33: Feature Code Listing Example

3.3 Annex C - Topographic Map Guidelines Annotation Catalog (TMS AC) Examples

This Annex provides the structure (with examples) required for an Annotation Catalog.

3.3.1 Overview

This Topographic Map Guidelines Annotation Catalog (TMS AC) defines example requirements for the portrayal of non-feature data (“metadata”) on a Topographic Map.

This Annex provides a framework guideline, structure of the sections and content required for establishing a Topographic Map Guidelines Annotation Catalog (TMS AC). This Annex serves as a container for Standard Operating Procedures (SOP’s) as and when these are created. Typical headings are listed and examples for 1:25,000 and 1:50,000 scale provided.

The content and examples are adopted from the National Geospatial Intelligence Agency (NGA) Standards.

For the full detail of the NGA Guideline refer to The NGA STANDARDIZATION DOCUMENT - DATA PRODUCT SPECIFICATION (DPS) - 1:25,000; 1:50,000 and 1:100,000 Scale Topographic Map (TM) - Annotation Catalog (AC) (2021-07-31) Version 1.2 (NGA.STND.0035-2_1.2_DPSTMAC).

3.3.2 Annotations within the Cartographic Process

Annotation is a graphic presentation of map display metadata (“non-feature data”) upon a cartographic product in order to provide necessary context for the display and comprehension of geographic features.

ISO 19117 defines the overall geographic portrayal process by which **features** are associated with **symbols** via the use of **portrayal rules** and **rule sets**. These symbols, rules and rule sets, along with additional process elements such as **generalization** and **finishing rules**, are contained in the Topographic Map Portrayal Catalog (TM PC) – see Annex B.

The concept of an Annotation extends the geographic portrayal process defined in ISO 19117. While **features** are associated with **symbols** in a Portrayal Catalog, **metadata** elements (about a data product) are associated with **annotation symbols** by the use of **annotation rules and rule sets**. The annotation process thus coexists with the geographic portrayal process within the larger cartographic process (see Figure 34).

Annotations take many forms depending on the contextual elements or metadata they communicate, as well as their placement within, and relationship to, the larger Map Display context.

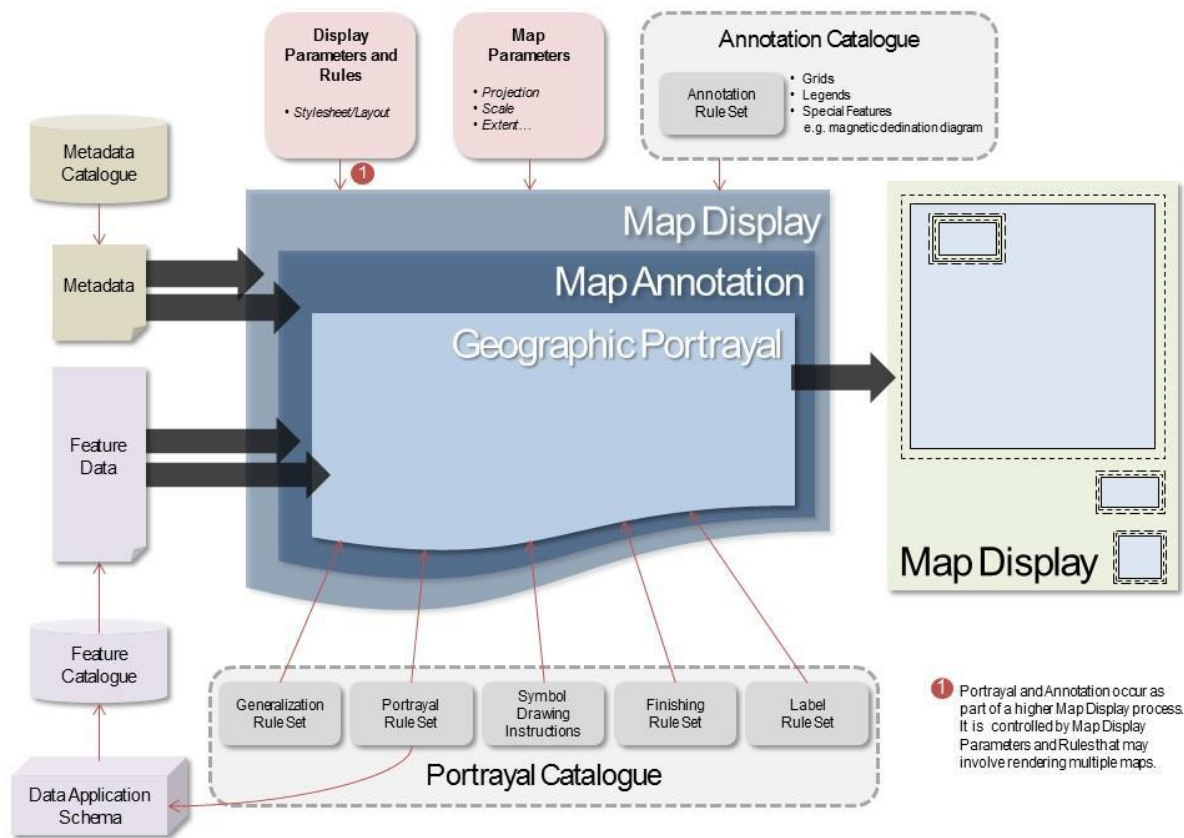


Figure 34: The cartographic process for a complete map display.

3.3.3 Basic Principles of Annotation

Information is included in the map margin which is necessary for map identification, feature identification, evaluation of map information and its intended purposes, identification of publishing elements, and releasability restrictions.

Content under this section shall cover the topics of Design & Location, Style & Font, Language

3.3.4 Annotation Guidance for Topographic Maps

This section presents guidance for constructing the annotations that are required or optional for Topographic Maps.

Content under this section shall cover the topics of Identification & Security Annotations (e.g. Map Sheet Name, Map Sheet Series Name, Publication Annotations, Security Markings), Georeferencing Annotations (e.g. Scale note & scale bar, Unit of elevation, elevation contour information notes) and Data Quality Annotations, Symbol Legends.

3.3.5 Annotation Catalog

The TM AC is organized into 2 sections:

- **Annotations:** presents all annotations from the TM AC. These annotations (along with portrayed feature data) constitute the basic components of a TM map display.
- **Style Sheets:** presents the graphic illustration and design of annotation elements and their locations on the map. Included in Annex B are subsections that provide more detailed versioning and information related to certain annotation elements.

3.3.5.1 Annotations

This presents all annotations from the TM DPS Annotation Catalog. These annotations (along with portrayed feature data) constitute the basic components of a TM map display.

An example of Annotation Record is shown below:

Symbol Legend

DPS AC Section: 3.5

Metadata Elements

Name:	Substitution Value:	Description:
Data Content - Maximum Elevation	dataContent.maxElevation	The maximum elevation value over the geographic extent of the map sheet.
Display Language	displayLanguage	The language(s) of presentation of the map sheet.

Annotation Rules

Reference #:	Description:	Criteria:
AC-0001	Always Include	TRUE

Symbol Legend (Single Language)

Style Sheet References:

1:25,000: Arrangement A
 1:50,000: Arrangement A
 1:100,000: Arrangement A

Version Notes:

(none)

Version Rules

Reference #:	Description:	Criteria:
AC-0002	Single-language	[displayLanguage.count] = 1

Components

Reference #: 200011	Name: Legend Title	Type: Text
Symbol Graphic Reference:	"LEGEND"	
Color:	Black	
Type Specification:	Zurich Condensed (9 point, Bold, Upper Case)	

Symbol Legend (Multiple Language)

Style Sheet References:

1:50,000: Arrangement B

Version Notes:

(none)

Version Rules

Reference #:	Description:	Criteria:
AC-0003	Multiple-language	[displayLanguage.count] > 1

Components

Reference #: 200011	Name: Legend Title	Type: Text
Symbol Graphic Reference:	"LEGEND"	
Color:	Black	
Type Specification:	Zurich Condensed (9 point, Bold, Upper Case)	
Reference #: 200013	Name: Legend Title (Translated)	Type: Text
Symbol Graphic Reference:	"LEGEND" (Translated)	
Color:	Black	
Type Specification:	Zurich Condensed (9 point, Medium, Upper Case)	
Reference #: 200012	Name: Legend Graphic	Type: Graphic

Figure 35: Example of Annotation Record

3.3.5.2 Annotation Rules

Rules consist of expressions involving metadata values that provide context or conditions and guide annotation appearance. Rules in the “Annotation Rules” section govern when the annotation will be shown within the TM map display. For each Annotation Rule, the following information is presented:

- The **reference number** of the rule (AC-XXXX)
- A **human-readable description** of the rule
- The **criteria** to be evaluated.
 - The criteria for each rule are a series of clauses that together evaluate to a Boolean value (TRUE/FALSE); an evaluation of TRUE indicates that the annotation will be shown, while FALSE means that it will not appear.
 - The rule criteria may take into account the value of the various metadata elements for the given TM. The metadata element’s substitution value, will be used as a placeholder for the actual value of the element.

Annotation Rules		
Reference #:	Description:	Criteria:
AC-0004	Classified	[sheetClassificationCode] is not 'unclassified'

Figure 36: Sample Annotation Rule

3.3.5.3 Annotation Version Rules

Version rules identify when a particular annotation version will be used and are listed after that version’s blue header (see Figure 37). The format and presentation of these rules matches that of the “Annotation Rules” as a whole.

If all of the criteria for a given Version Rule evaluate to TRUE for a TM instance, that particular version of the annotation should be used for the instance. Version rules are constructed so that only one version of each annotation will evaluate to TRUE in this way.

If there is only a single version of an annotation (i.e., there are no blue headers for separate annotation versions), the Version Rules section will not appear – the Annotation Rules alone govern the appearance of the annotation.

Version Rules		
Reference #:	Description:	Criteria:
AC-0007	1:50K	[displayScale.equivalentScale] = 1:50,000
AC-0003	Multiple-language	[displayLanguage.count] > 1

Figure 37: Example Annotation version rule.

3.3.5.4 Annotation Components

The Components section describes in detail the components that make up the annotation version. Like feature symbols, each annotation is made up of multiple graphical (lines and other geometric symbols) and textual elements. Components are not inherently annotation-specific: they may be reused among annotations or annotation versions, although most components are only used in one annotation/version.

Figure 38 provides an example of a complete components section. For each component (regardless of type), the following characteristics are listed:

- The full **reference number** of the component, a 6-digit number that uniquely identifies the component regardless of the annotation/version it is a part of;
- The descriptive **name** of the component; and
- The **type** of the component (text, graphic, map, or other); (at bottom) Additional **notes** on the component (if needed).

Components

Reference #: 200411	Name: Ellipsoid Note	Type: Text
Symbol Graphic Reference:	"ELLIPSOID...WORLD GEODETIC SYSTEM 1984"	
Color:	Black	
Type Specification:	Zurich Condensed (6 point, Medium, Upper Case)	
Note:	The ellipsoid used for all maps is the World Geodetic System 1984.	
Reference #: 200412	Name: Grid Note (Single)	Type: Text
Symbol Graphic Reference:	"GRID...1,000 METER UTM ZONE [primaryMapMajorGridZone]"	
Color:	Black	
Type Specification:	Zurich Condensed (6 point, Medium, Upper Case)	
Note:	The grid system used for all maps is UTM.	
Reference #: 200413	Name: Projection Note	Type: Text
Symbol Graphic Reference:	"PROJECTION...TRANSVERSE MERCATOR"	
Color:	Black	
Type Specification:	Zurich Condensed (6 point, Medium, Upper Case)	
Note:	The projection for all maps is Transverse Mercator.	
Reference #: 200414	Name: Vertical Datum Note	Type: Text
Symbol Graphic Reference:	"VERTICAL DATUM...MEAN SEA LEVEL"	
Color:	Black	
Type Specification:	Zurich Condensed (6 point, Medium, Upper Case)	
Note:	The vertical datum for all maps is Mean Sea Level.	
Reference #: 200415	Name: Horizontal Datum Note	Type: Text
Symbol Graphic Reference:	"HORIZONTAL DATUM...WORLD GEODETIC SYSTEM 1984"	
Color:	Black	
Type Specification:	Zurich Condensed (6 point, Medium, Upper Case)	
Note:	The horizontal datum for all maps is the World Geodetic System 1984.	

Figure 38: Annotation components section example.

3.3.5.5 Style Sheets

Style Sheets are included to provide a graphical layout of the map sheets and serve as a template for visual understanding of the make-up of the sheets and the placement of the various cartographic components such as annotation and symbology based on the rules and norms detailed in the Annotation and Portrayal Catalogs.

Example Style Sheets and map sheet graphics are provided in **Annex D**

- 1:50,000 Scale Topographic Maps Style Sheet Arrangement “A” – single language/single grid
- 1:50,000 Scale Topographic Maps Style Sheet, Arrangement “B” – multiple languages/overlapping grid
- 1:25,000 Scale Topographic Maps Style Sheet, Arrangement “A” – single language/single grid
- Master Legend: illustrates the design and composition of a standard symbol legend to be used on a TM.
- Grid Declination Diagrams: provides detailed guidelines pertaining to the composition and portrayal of declination diagram arrangements.
- Margin Symbol Detailed Drawing Instructions: provides detailed drawing instructions of unique marginalia symbols (road designation and box arrow, grid declination star, grid declination full and half arrow).
- Slope Guides
- Grid Zone Junctions

3.4 Annex D – Topographic Map Guidelines Style Sheet (TMS SS) Examples

This Annex provides examples of topographic map Style Sheets

- Example of 1:25,000-Scale Topographic Map Style Sheet
- Example of 1:50,000-Scale Topographic Map Style Sheet

3.4.1 Overview

This Annex provides Topographic Map Guidelines STYLE SHEET (TMS SS) to accompany Annex B -Topographic Map Guidelines Portrayal Catalog (TMS PC) and Annex C - Annotation Catalog (TMS AC) of the KSA Topo-Map Guidelines for Topographic Maps.

This Annex provides a framework guideline, structure of the sections and content required for establishing a Topographic Map Guidelines Style Sheets (TMS SS). This Annex serves as a container for Standard Operating Procedures (SOP's) as and when these are created. Typical headings are listed and examples for 1:25,000 and 1:50,000 scale provided.

Due to the size of the example map sheets, some content is presented at low resolution. The full resolution examples can be accessed from separate full resolution PDF files with links provided.

The content and examples are adopted from the National Geospatial Intelligence Agency (NGA) Standards.

For the full detail of the NGA Guideline refer to NGA STANDARDIZATION DOCUMENT - DATA PRODUCT SPECIFICATION (DPS) - 1:25,000; 1:50,000 and 1:100,000 Scale Topographic Map (TM) - Annotation Catalog (AC) (2021-07-31) Version 1.2 (NGA.STND.0035-2_1.2_DPSTMAC).

3.4.2 Style Sheets and supporting sections

Example Style Sheets and map sheet graphics are provided.

- 1:50,000 Scale Topographic Maps Style Sheet Arrangement "A"
- 1:50,000 Scale Topographic Maps Style Sheet, Arrangement "B"
- 1:25,000 Scale Topographic Maps Style Sheet, Arrangement "A"
- Master Legend
- Grid Declination Diagrams
- Margin Symbol Detailed Drawing Instructions
- Slope Guides
- Grid Zone Junctions

3.4.2.1 Example 1:50 000 Scale Topographic Maps Style Sheet Arrangement "A"

see separate PDF for full resolution version of this example - [TM50_Arrangement_A_v1.2_JUL21.PDF](#)

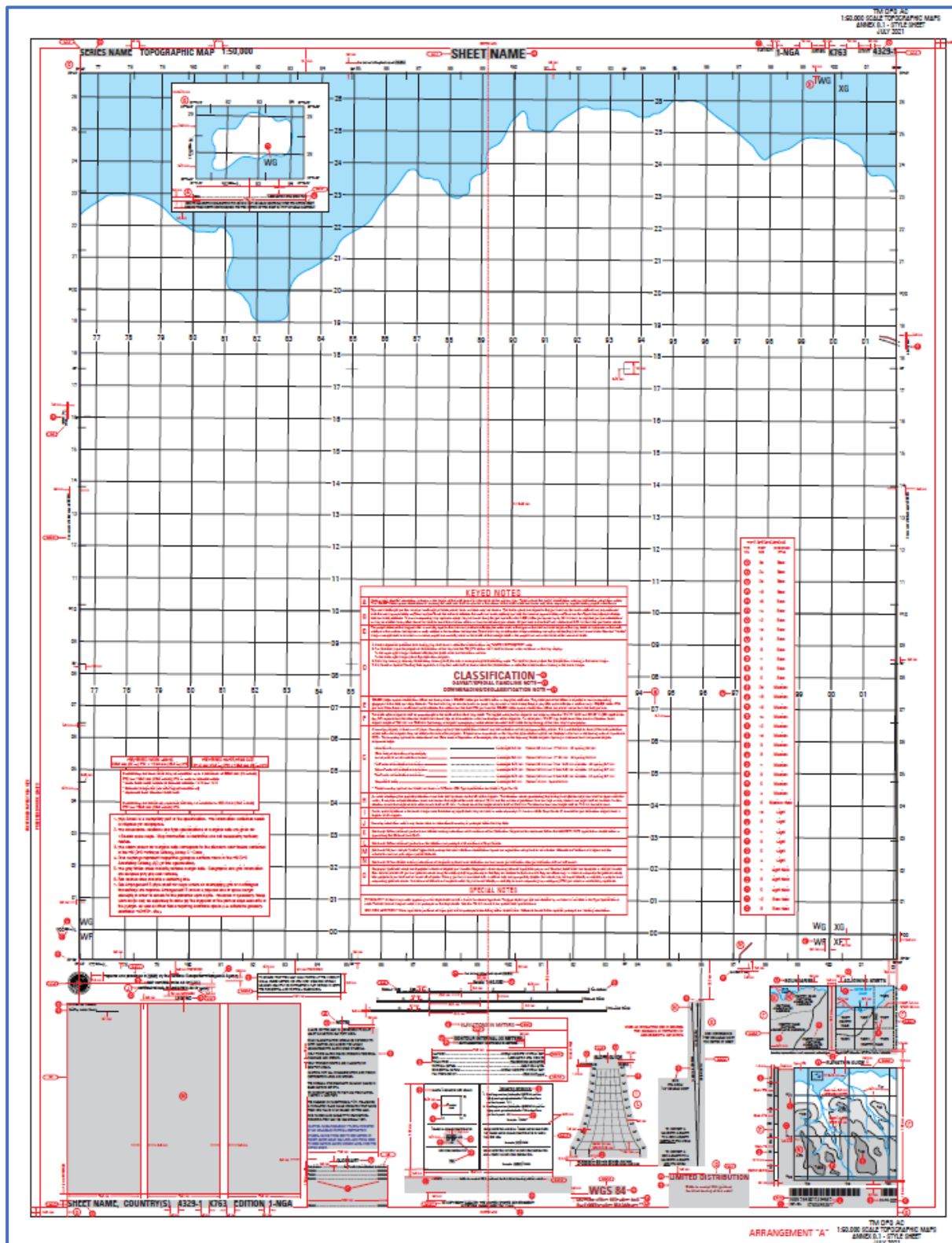


Figure 39: Example 1:50 000 Scale Topographic Maps Style Sheet Arrangement "A"

3.4.2.2 Example 1:50 000 Scale Topographic Maps Style Sheet Arrangement “B”

see separate PDF for full resolution version of this example - *TM50_Arrangement_B_v1.2_JUL21.PDF*

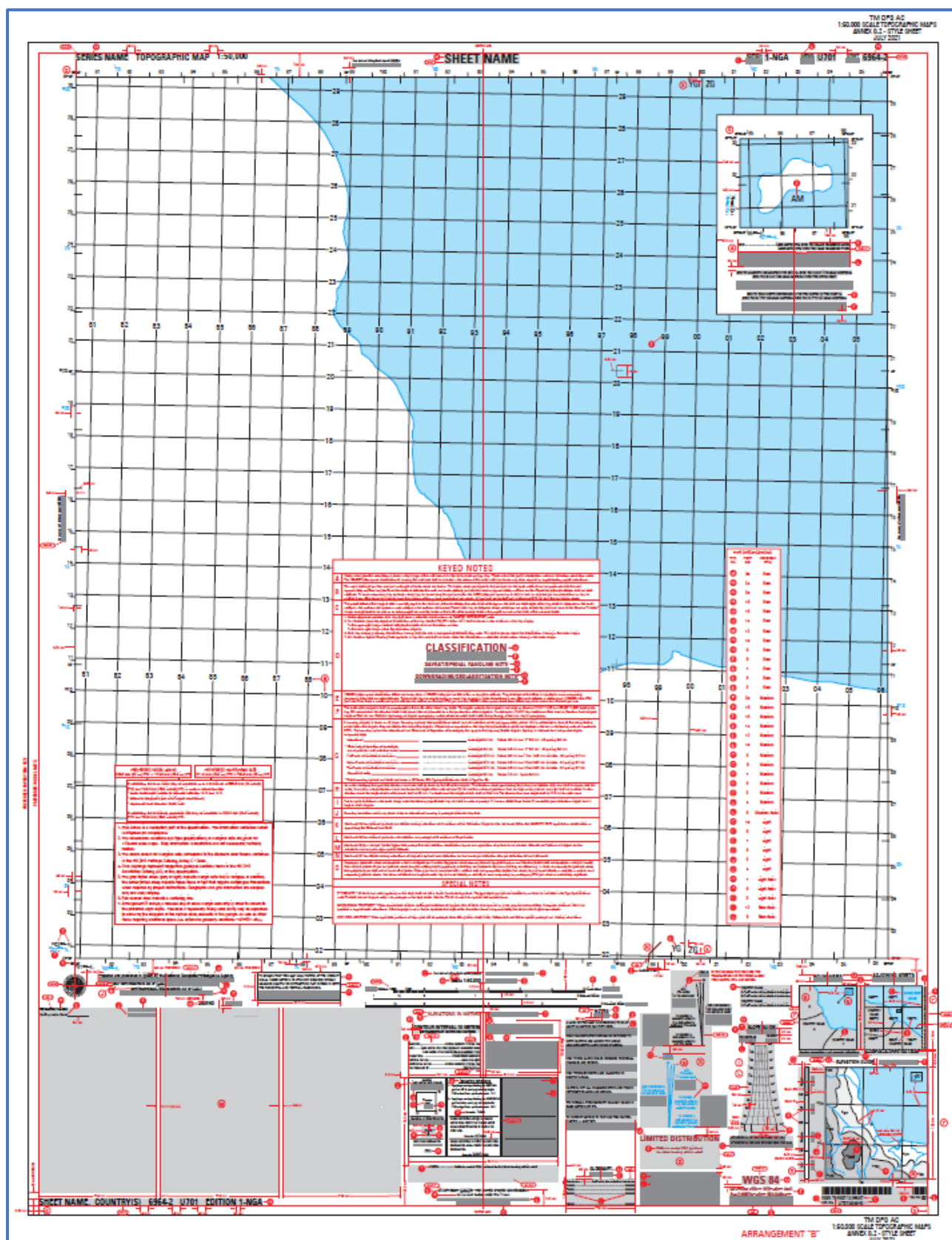


Figure 40: Example 1:50 000 Scale Topographic Maps Style Sheet Arrangement “B”

3.4.2.3 Example 1:25 000 Scale Topographic Maps Style Sheet Arrangement "A"

see separate PDF for full resolution version of this example - [TM25_Arrangement_A_v1.2_JUL21.PDF](#)

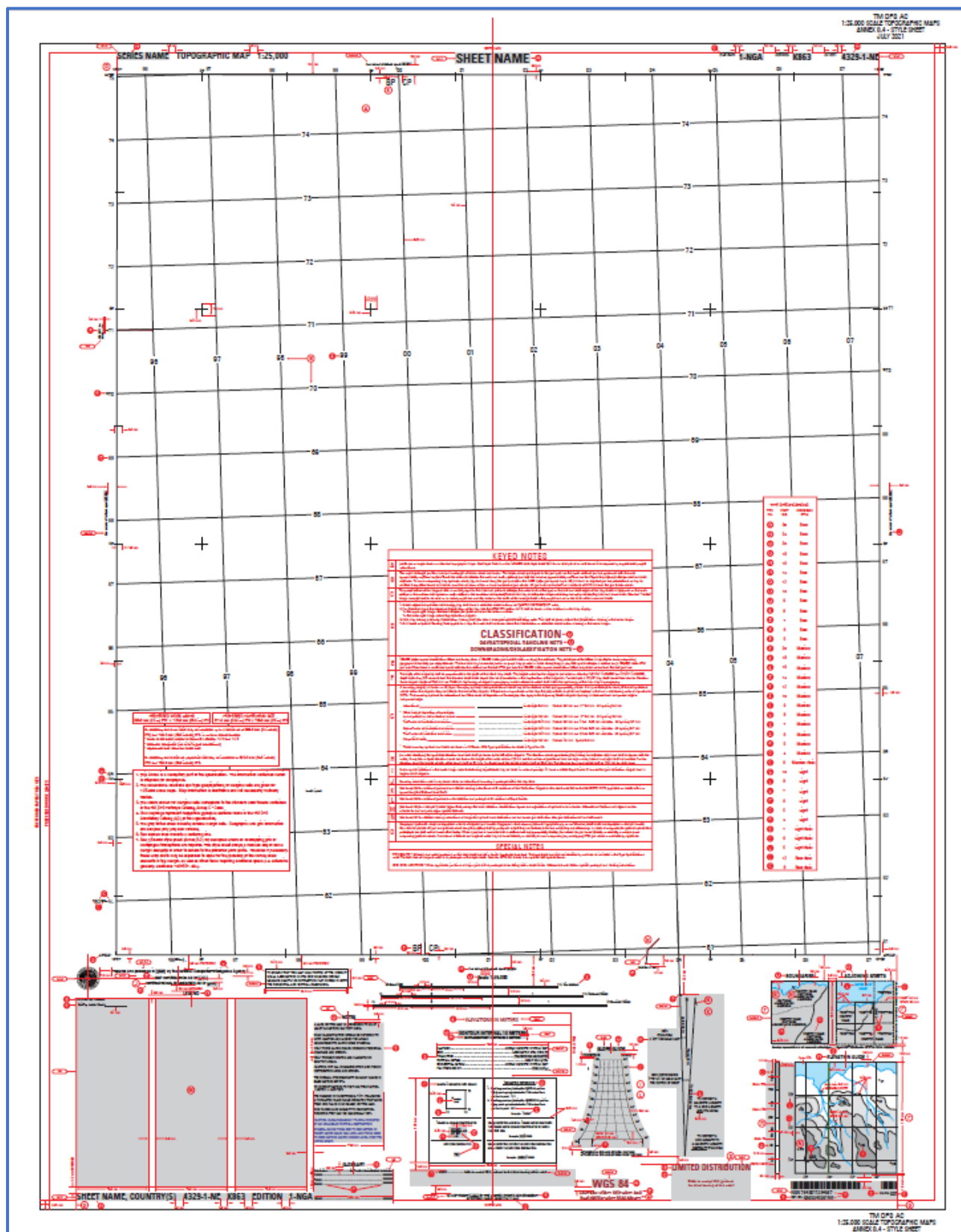


Figure 41: Example 1:25 000 Scale Topographic Maps Style Sheet Arrangement "A"

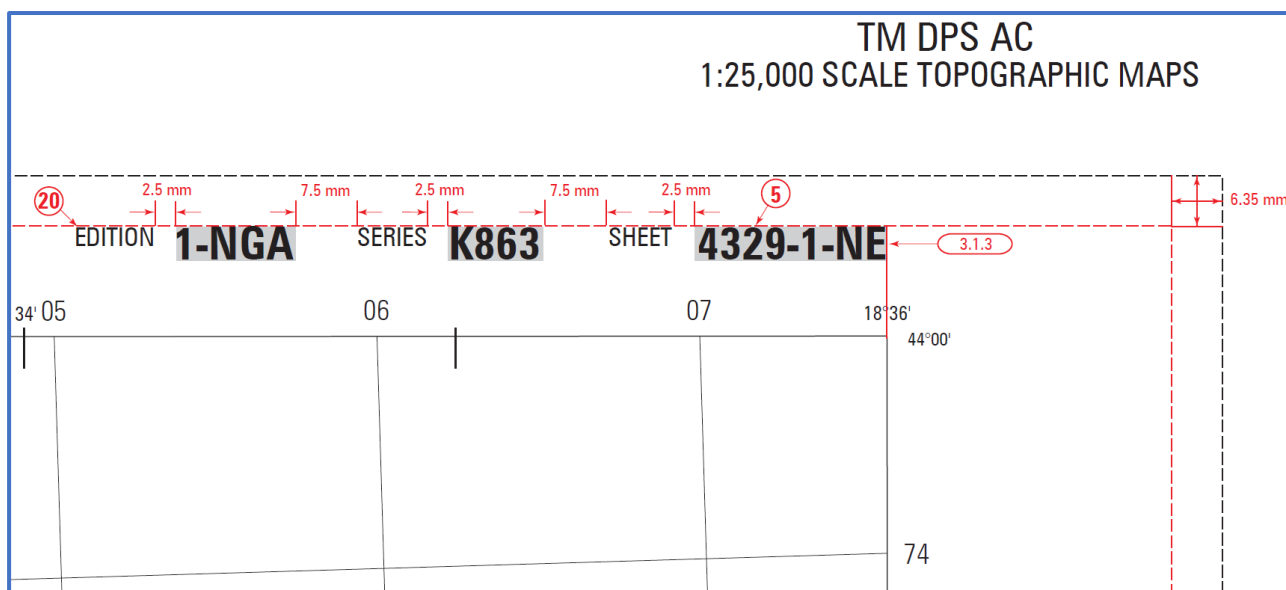


Figure 42: Example enlarged detail - 1:25 000 Scale Topographic Maps Style Sheet Arrangement "A"

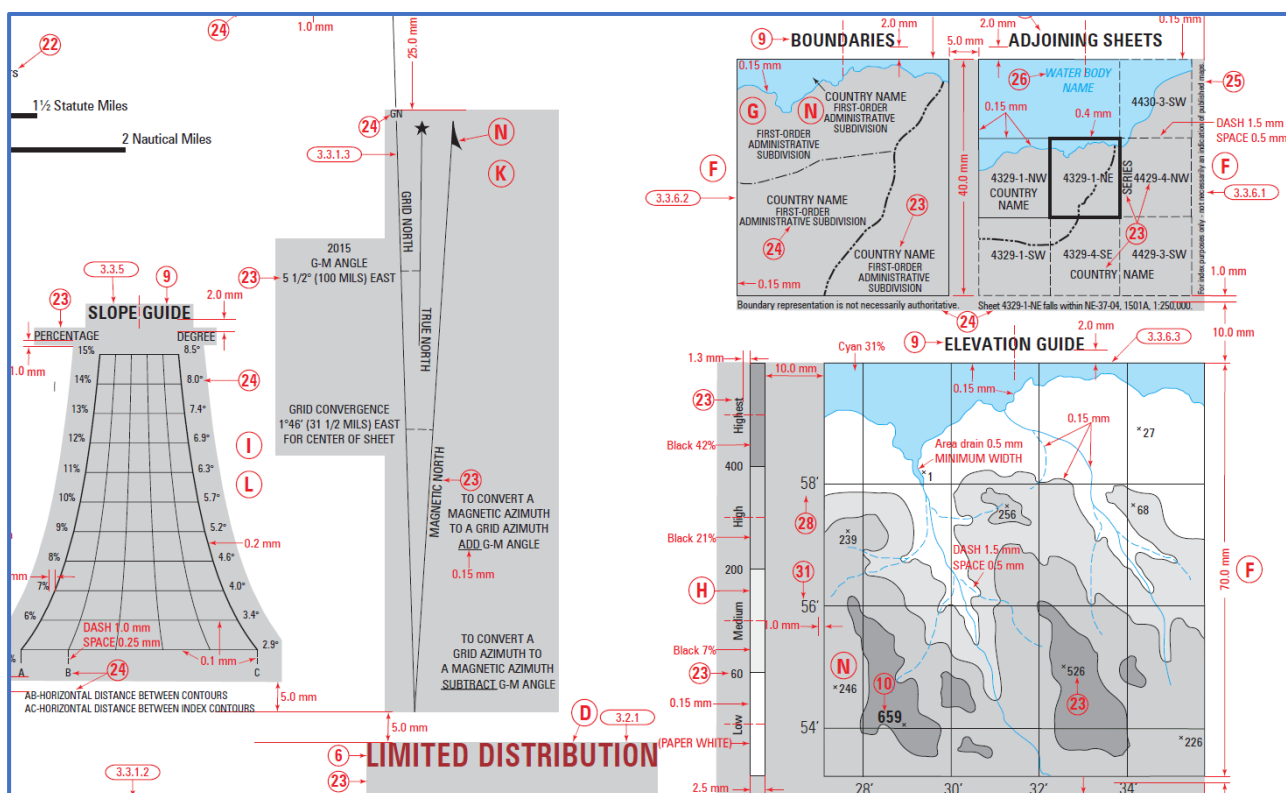


Figure 43: Example enlarged detail - 1:25 000 Scale Topographic Maps Style Sheet Arrangement "A"

3.4.2.4 Example Master Legend

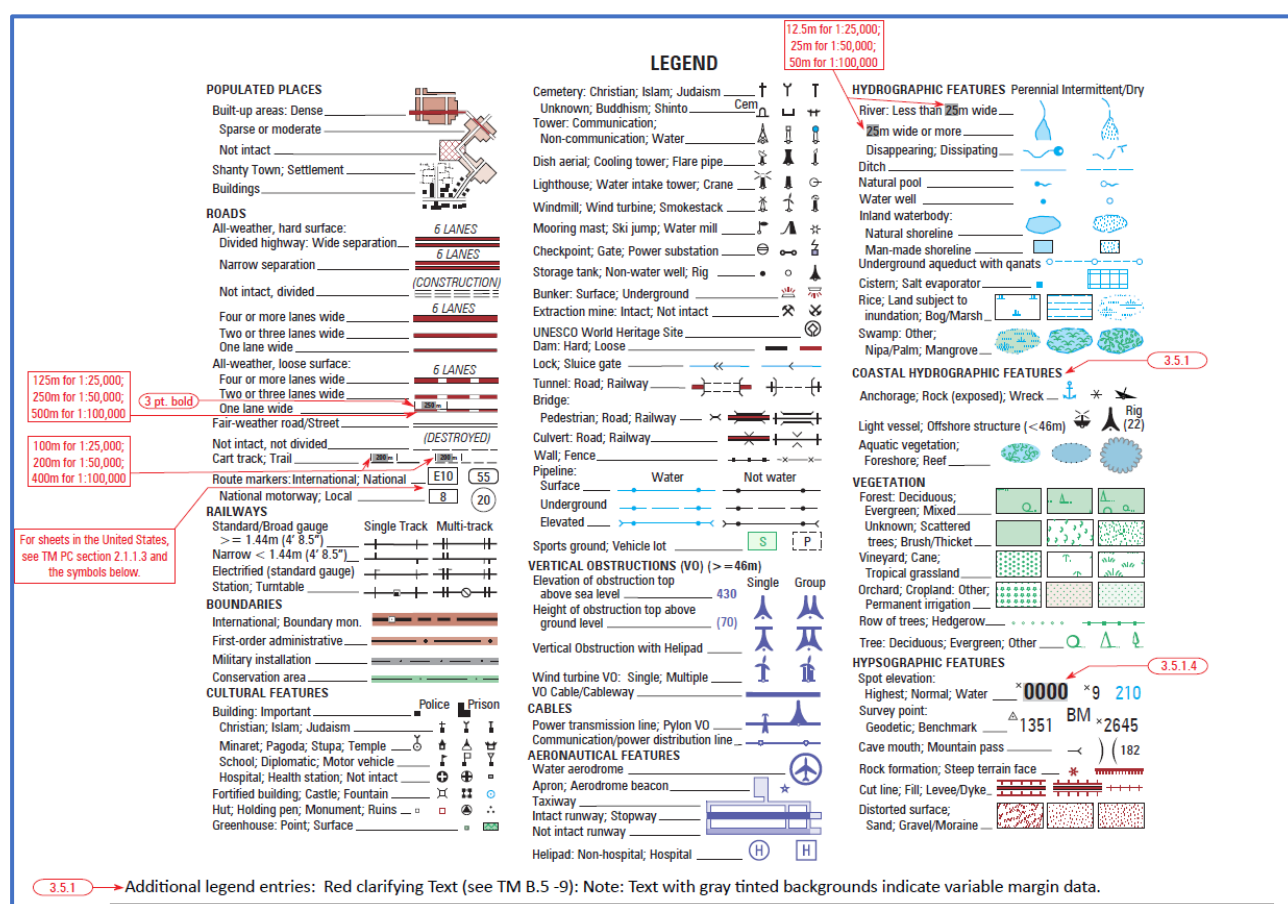


Figure 44: Example Master Legend

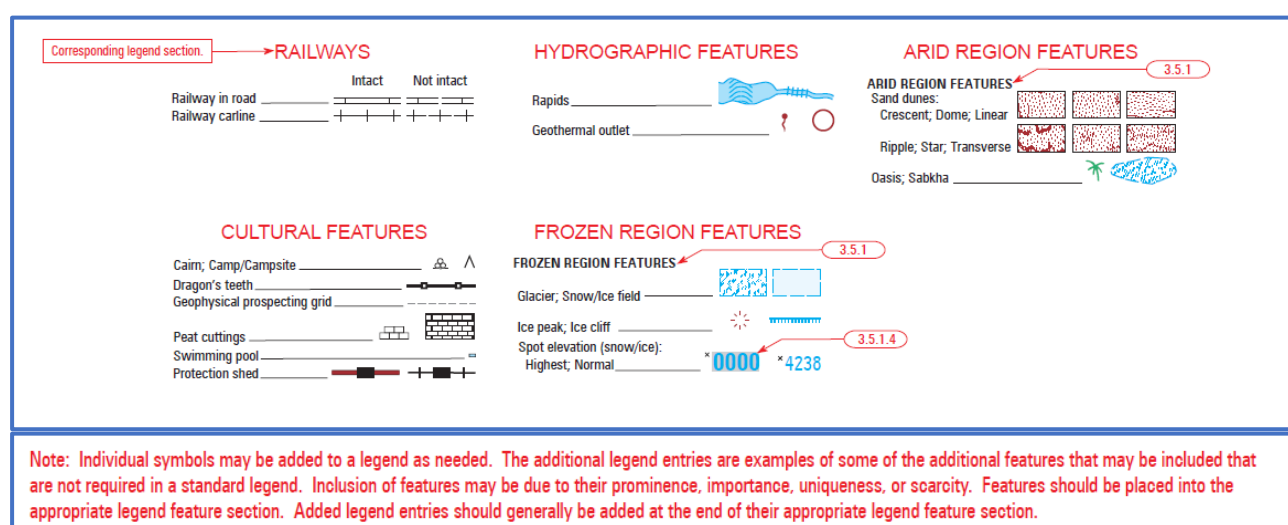


Figure 45: Additional Details on the Master Legend Example

3.4.2.5 Example Grid Declination Diagrams

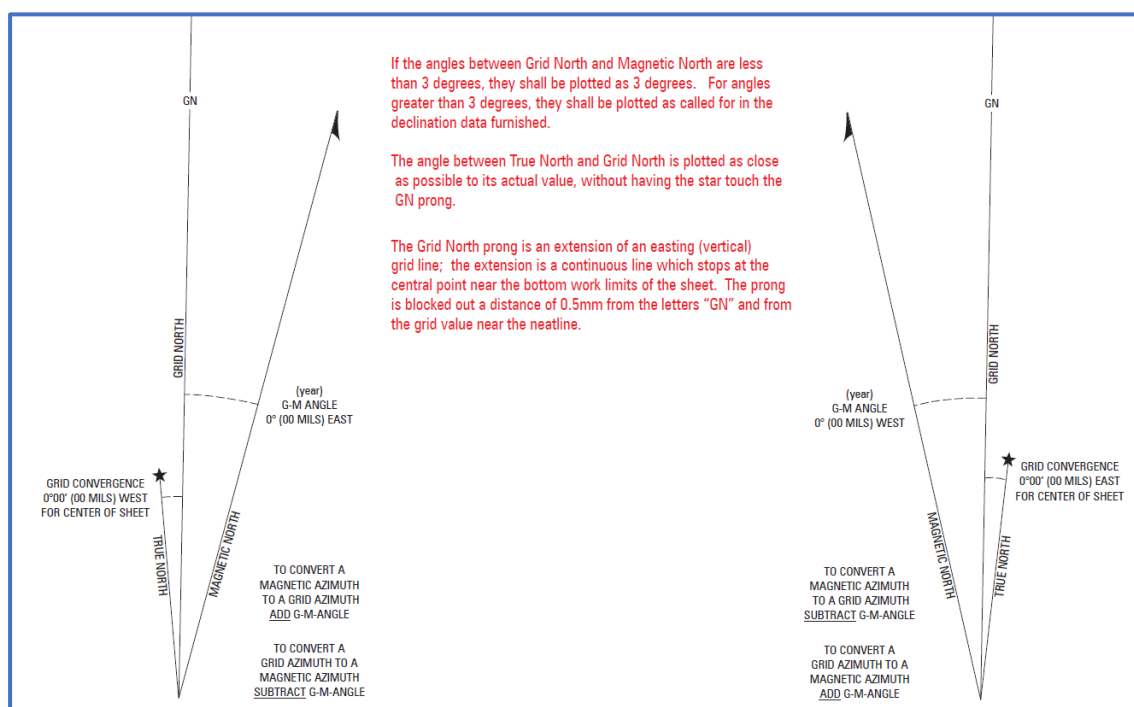


Figure 46: Example-1 Grid Declination Diagrams

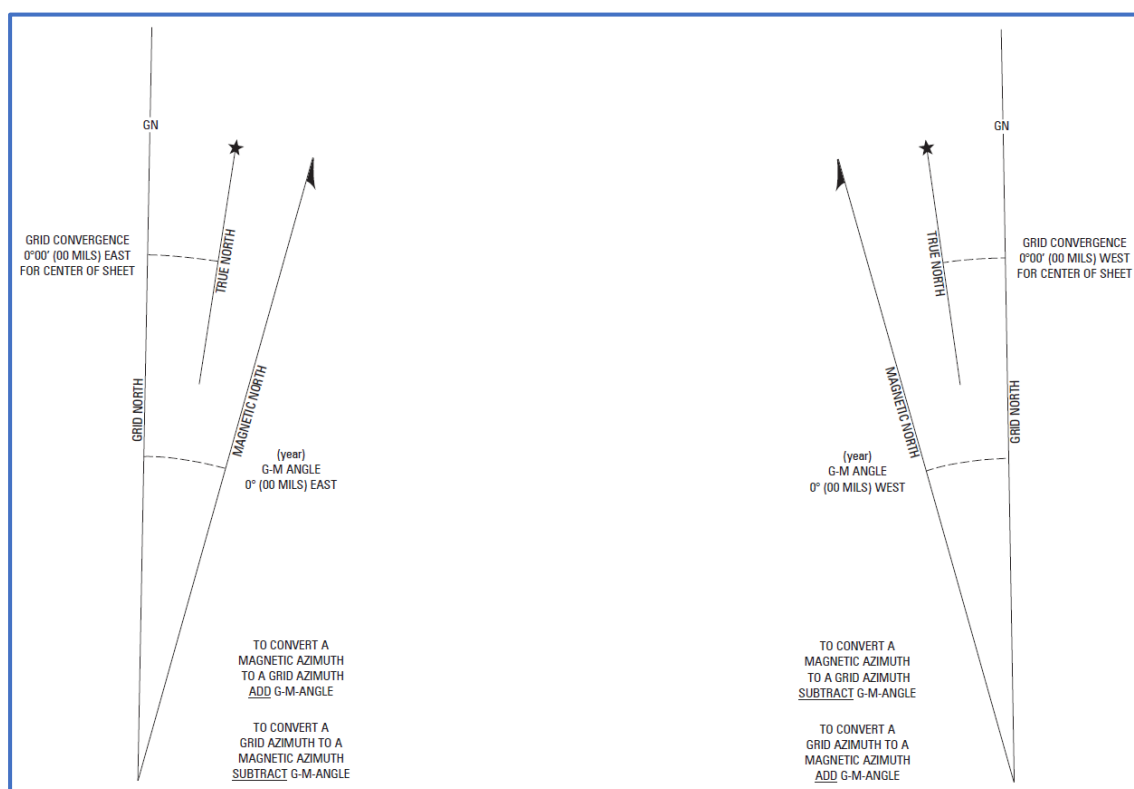


Figure 47: Example-2 Grid Declination Diagrams

3.4.2.6 Example Margin Symbol Detailed Drawing Instructions

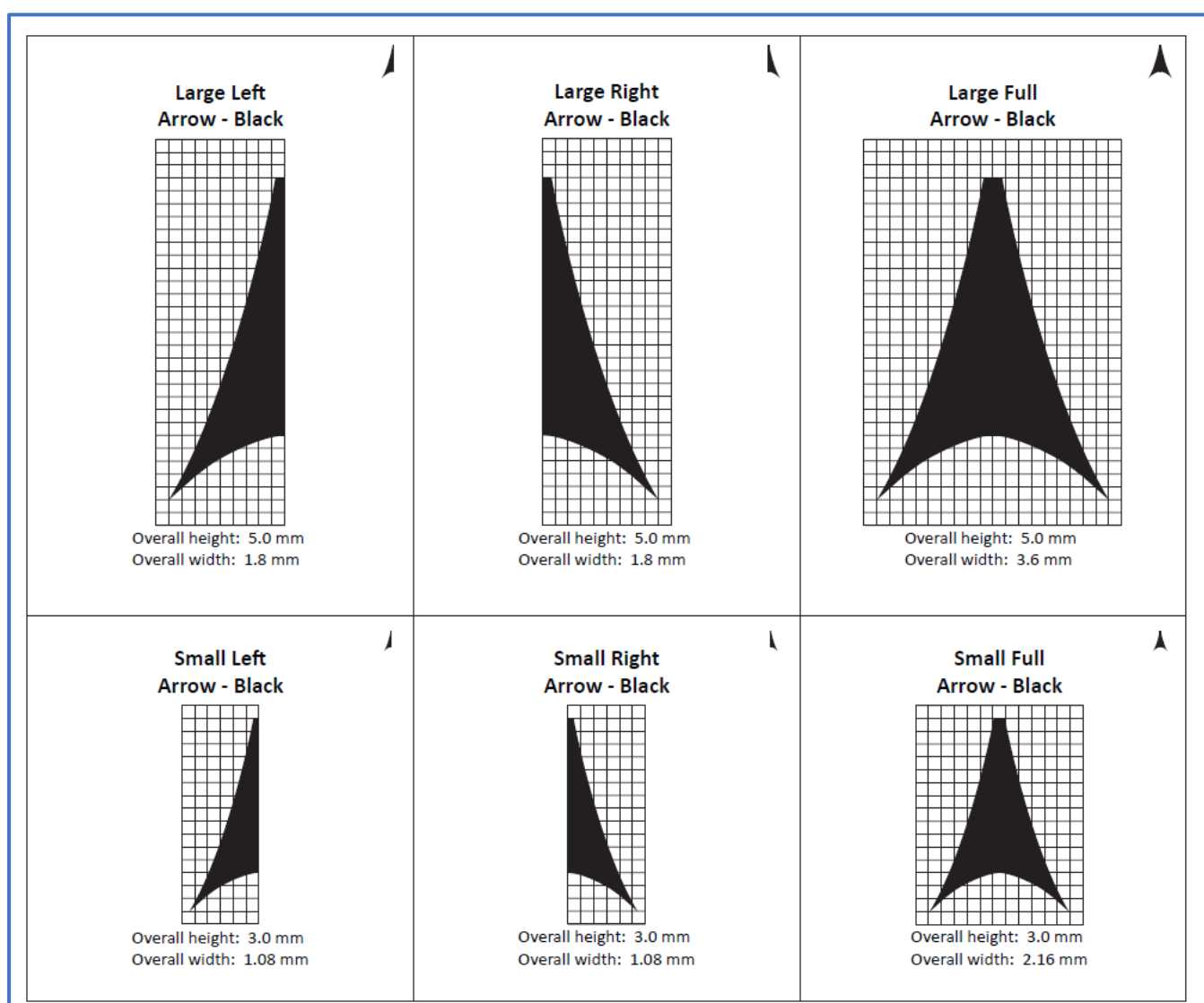


Figure 48: Example-1 Margin Symbol Detailed Drawing Instructions

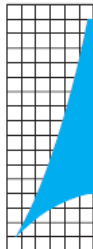
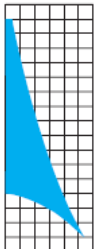
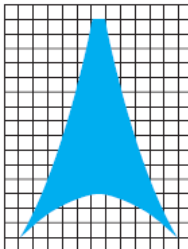
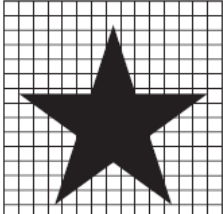
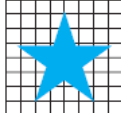
Small Left Arrow - Cyan  Overall height: 3.0 mm Overall width: 1.08 mm	Small Right Arrow - Cyan  Overall height: 3.0 mm Overall width: 1.08 mm	Small Full Arrow - Cyan  Overall height: 3.0 mm Overall width: 2.16 mm
Large Star - Black  Overall height: 2.472 mm Overall width: 2.6 mm	Small Star - Cyan  Overall height: 1.236 mm Overall width: 1.3 mm	Small Star - Cyan  Overall height: 1.236 mm Overall width: 1.3 mm
Elevation Guide Spot Elevation  Lineweight: 0.1 mm Length of lines: 0.8 mm Notes: Lines intersect at 45 degree angles to form an "X".	Road Destination and Diagrams Arrow  Arrow staff lineweight: 0.2 mm Overall height: 1.0 mm Overall length: 6.0 mm (Length of diagram arrows may be adjusted as needed)	

Figure 49: Example-2 Margin Symbol Detailed Drawing Instructions

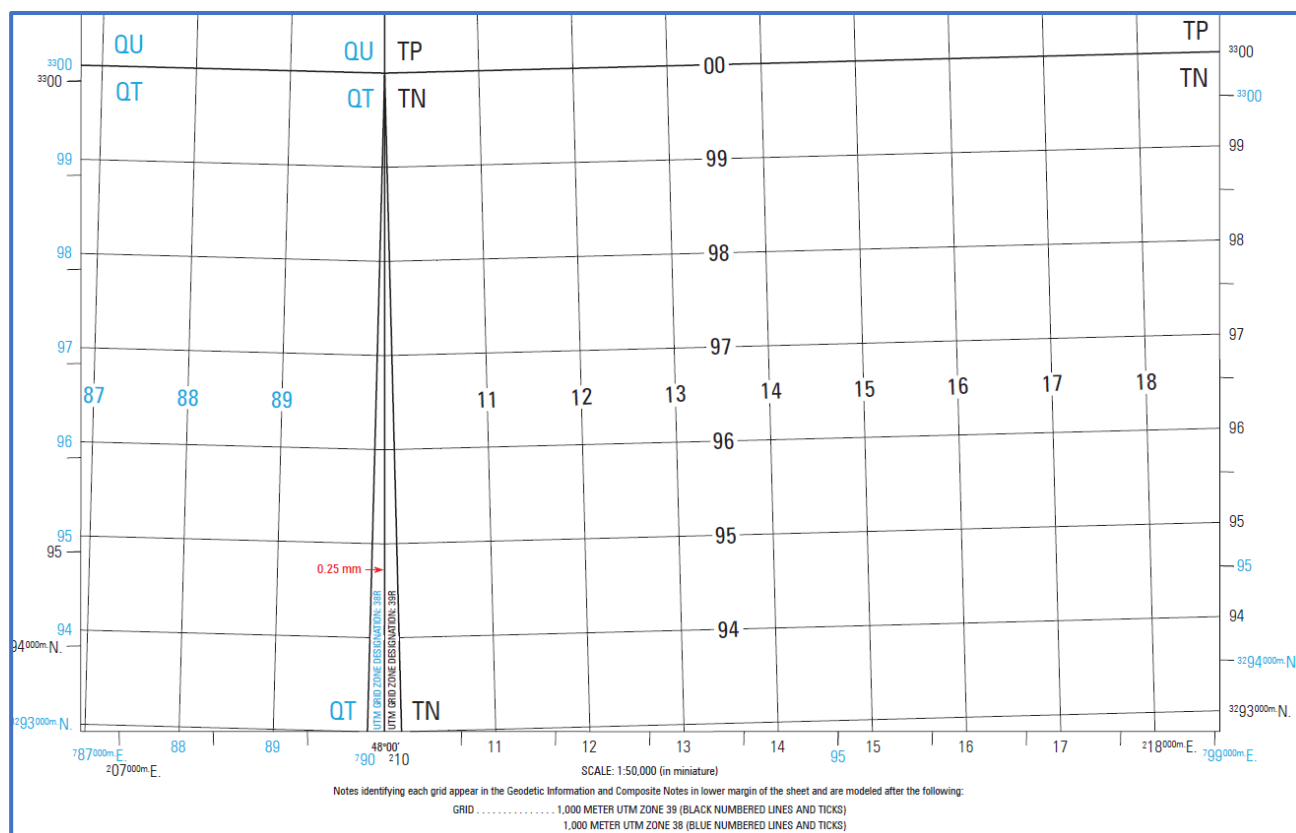


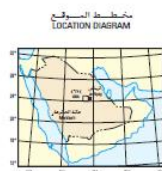
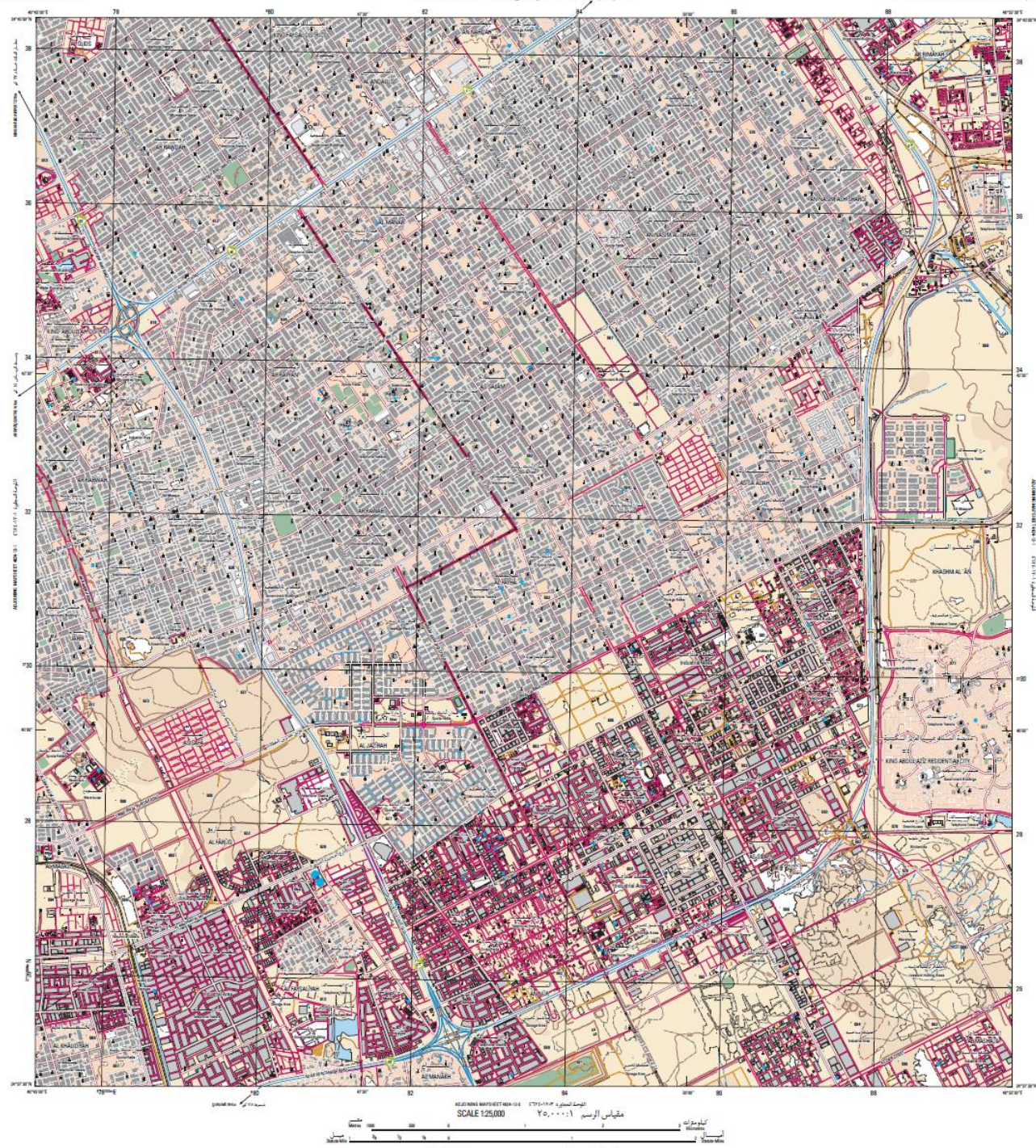
Figure 52: Example-2 Grid Zone Junctions

3.5 Annex E – Example 1:25 000 Topographic Map

The 1:25,000 Scale KSA topographic map, with civilian applications, which includes information on urban populations, isolated and rural buildings, transportation networks and utilities, road distances, and route destinations, telecommunications networks and facilities, power supply grids and facilities, industrial works, farmland and plantations, water supply and wastewater utilities, vegetation, surface natural features and relief of the coverage area, heights above sea level in meters depicted by contour lines, spot elevations, and color shades, place names, landmarks, natural and cultural features.

As part of the KSA national map series, the map contains Arabic and English text.

The full resolution map sheet is contained in a separate PDF document [4624124_Riyadh_SE1 Copy.pdf](#)



مخطط الموقع
LOCATION DIAGRAM

شرح المفردات

[illegible]

ت الإنتاج

[illegible]

LEGEND

[illegible]

مصطلحات الخارطة

[illegible]

معلومات الحضانة

74 68



خط: عار كتر المستوطن العالمي
A. النظام الاقتصادي العالمي

تتولى صناعات مطبخ البحر (شركة ١٩٦٦م)

PROJECTION AND GRID
Projection: Universal Transverse Mercator
Spheroid: WGS 84
Horizontal Datum: WGS 84
Vertical Datum: Mean Sea Level (MSL)

مطابق الخريطة على الإسكندرية كور
من تصفحات كور البحر المتوسط والخليج
٣٨، ارتفاع المتوسطي البحري (MSL)

Unpublished 3rd Issue Indicated
3,000 Meters Universal Transverse Mercator
Zone 36, World Geodetic System



الهيئة العامة للمساحة والمعلومات الجيوعكانية
المملكة العربية السعودية

General Authority for Survey
and Geospatial Information
Kingdom of Saudi Arabia

الرياض (جنوب غرب)
AR RIYĀD (SOUTH EAST)
منطقة الرياض
AR RIYĀD REGION

3.6 Symbols, Acronyms and Abbreviations

Symbols, Acronyms and Abbreviations	Definition
AC	Annotation Catalog
AS	Application Schema
CADRG	Compressed Arc Digitized Raster Graphic
CMYK	Cyan, Magenta, Yellow, Black
DPS	Data Product Specification
ECRG	Enhanced Compressed Raster Graphic
FC	Feature Catalog
GEOSA	General Authority for Survey and Geospatial Information
GML	Geography Markup Language
MGCP	Multinational Geospatial Co-production Program
MGRS	Military Grid Reference System
MSL	Mean Sea Level
MXD	Map Document Files format from ESRI
NFCD	National Feature Concept Dictionary
PC	Portrayal Catalog
PDF	Portable Document Format
RN	Reference Number
SANSRS	Saudi Arabia National Spatial Reference System
SOPs	Standard Operating Procedures
TDS	Topographic Data Store
TDS EC	Topographic Data Store Entity Catalog
TDS EG	Topographic Data Store Extraction Guide
TM	Topographic Map
TMS	Topographic Map Specification
TMS AC	Topographic Map Specification Annotation Catalog
TMS FC	Topographic Map Specification Feature Catalog
TMS PC	Topographic Map Specification Portrayal Catalog

Symbols, Acronyms and Abbreviations	Definition
UTM	Universal Transverse Mercator
XML	Extensible Markup Language

3.7 References

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